Tidal Power

Definition and Motivation

- What is Tidal Energy?
  - Energy contained in a tide, especially that which can be converted to electricity.
- Why Tidal Energy?
  - Declining availability of fossil fuels and global climate change.

Tidal Basics

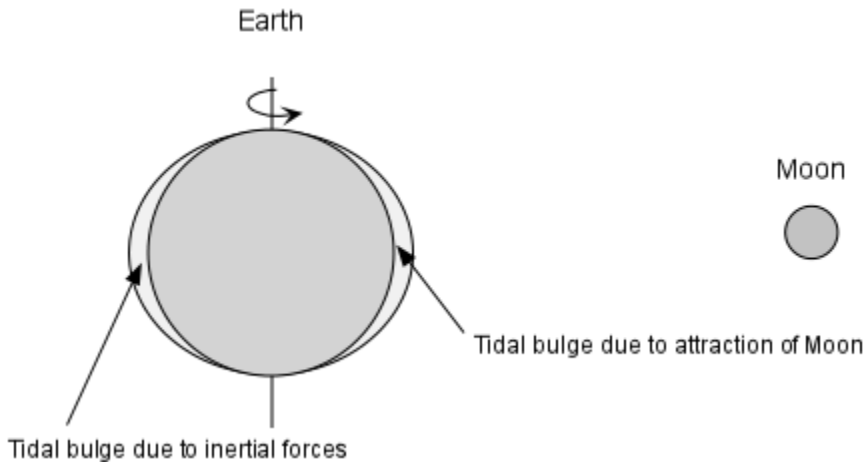
- Most locations have two tidal cycles per day: 12 hours, 25 minutes
- Essentially caused by interaction of moon, earth, and sun centrifugal forces
- Diurnal tides are generated because the maxima and minima in each daily rotation are unequal in amplitude

Particle (Water)

Radius of Earth

$$\text{Tidal Force} = \frac{2Gmm_1a}{R^3}$$

Distance between Earth and Moon



Newton's Law of Gravitational State

$$F = G\frac{m1m2}{R^2}$$

Tidal Facts

- Tidal Range = Distance between high tide mark and low tide mark
- Largest Ranges→ Coastal Regions with extreme depth gradient (Mouzel = excellent tidal range)
- Smallest Ranges→ Open Ocean (<.5 m)

Spring Tides

- Year round
- Occur during full moon and new moon
- Due to the linear pattern of SME
- Causes stronger tides: increased current and tidal ranges

Gravitational Effect of the Sun

a. **Spring Tides**

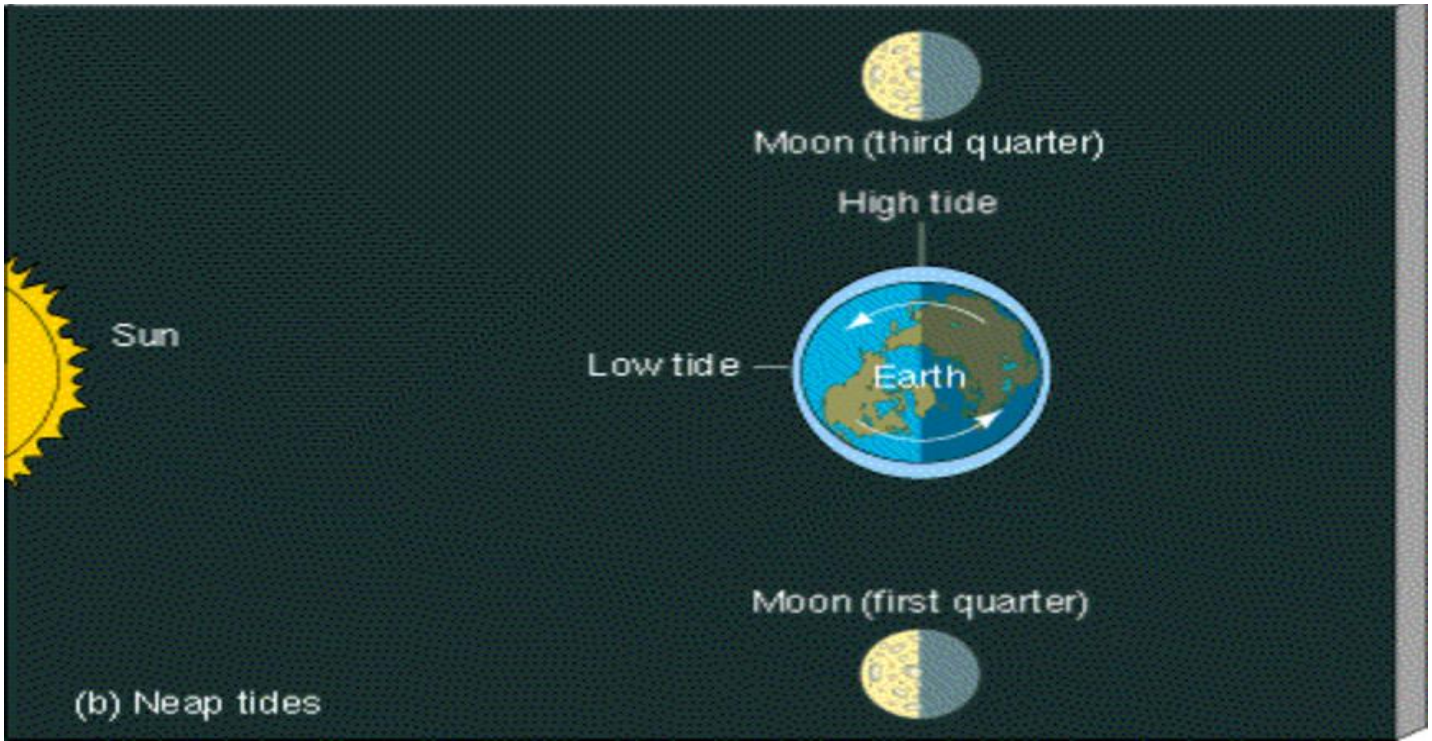
- Earth, Moon, and Sun are lined up and work together
- High Tides are higher and Low Tides are lower than normal

Neap Tides

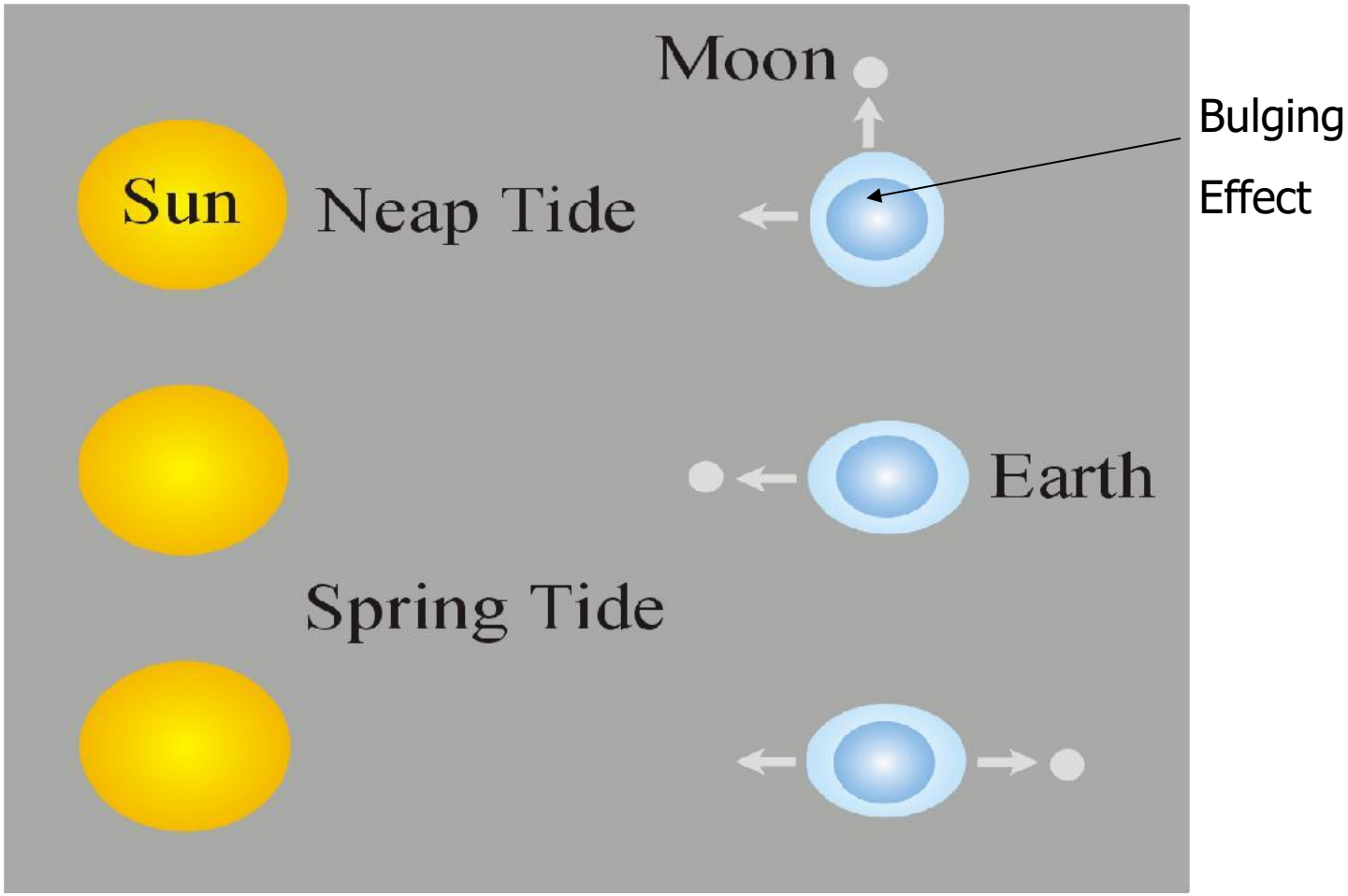
- Moon and sun are perpendicular to each other
- Weak currents, lower tidal ranges
- Occur during quarter moons

b. Neap Tides

- Earth, Moon, and Sun form right angles and work against each other
- High Tides are lower and Low Tides are higher than normal

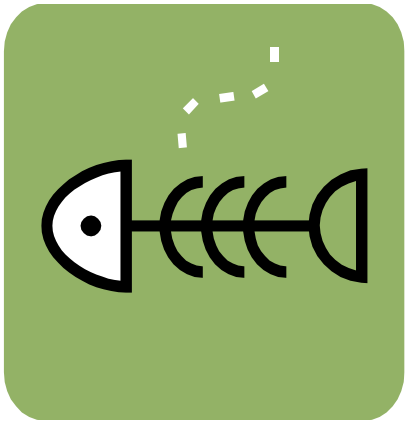


Neap Tides Versus Spring Tides



Disadvantages

- Environmental



- Economical

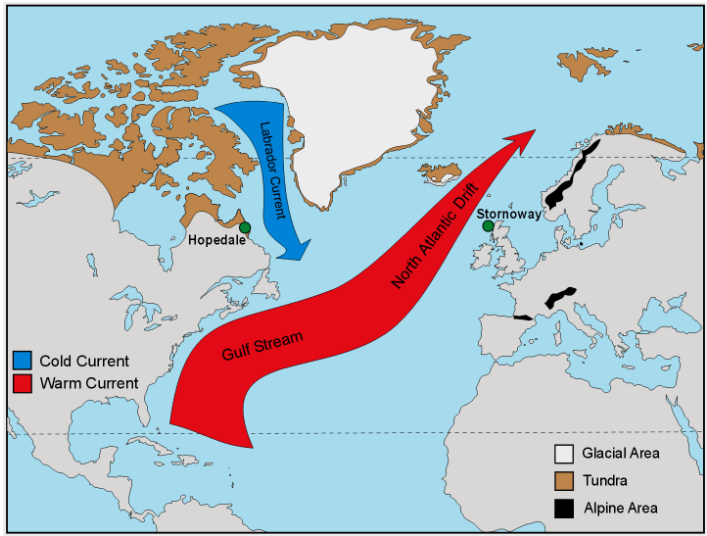
- Design

ocean currents

A. Ocean water contains horizontal, stream-like movements of water called *ocean currents*.
B. Affected by weather, Earth's rotation, and the position of the continents.
C. Importance:
   1. moves drifting organisms from place to place – plankton, disperse young
   2. carries eggs and larvae of organisms that have external fertilization
   3. brings food, oxygen
   4. carries away waste, pollutants

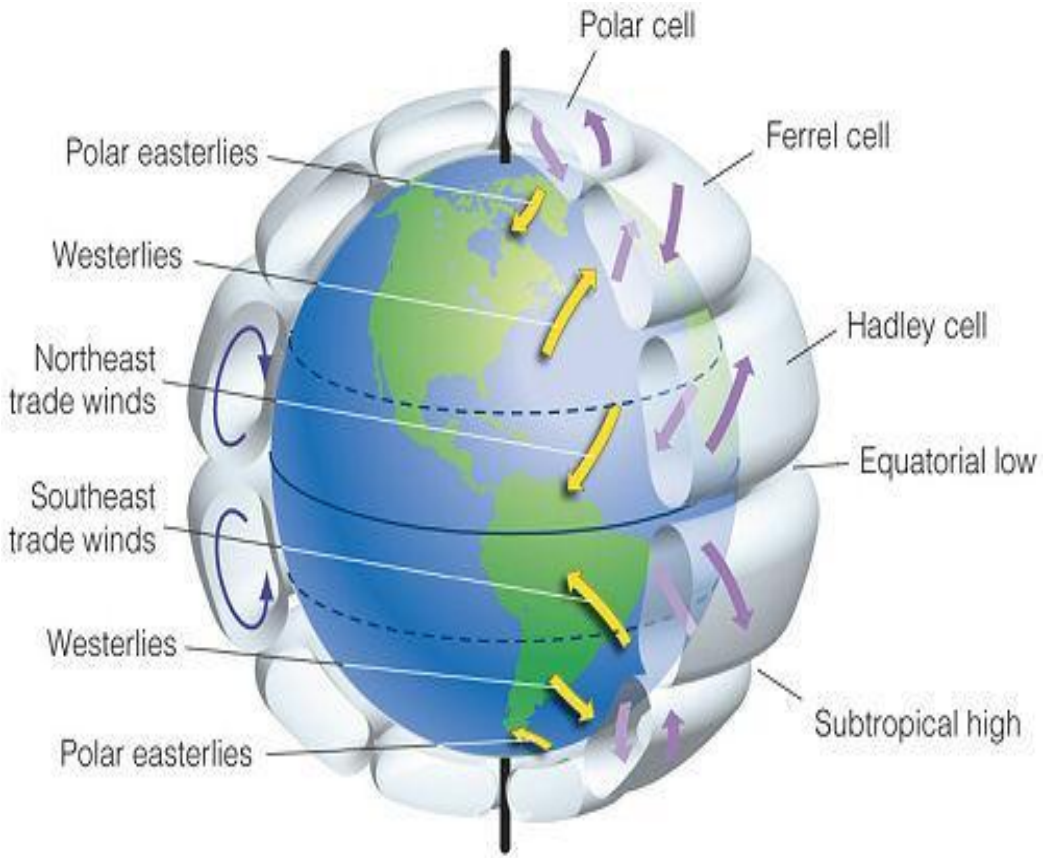
Three Main Types

1. Surface Currents
   a. Horizontal movements of ocean water caused by wind and occurring at or near the ocean's surface are called surface currents.
   b. Can reach depths of several hundred meters and lengths of several thousand kilometers.
   c. The Gulf Stream is one of the longest surface currents, transporting 25 times more water than all the rivers in the world combined.
   d. Controlled by 3 factors:
      - » Global winds
      - » Continental barriers
      - » Coriolis Effect



Global Winds

- Different winds cause currents to flow in different directions.
  - The *trade winds* are located just north and south of the equator.
    - In both hemispheres, they push currents westward across the tropical latitudes.
  - The *westerlies* are located in the middle latitudes.

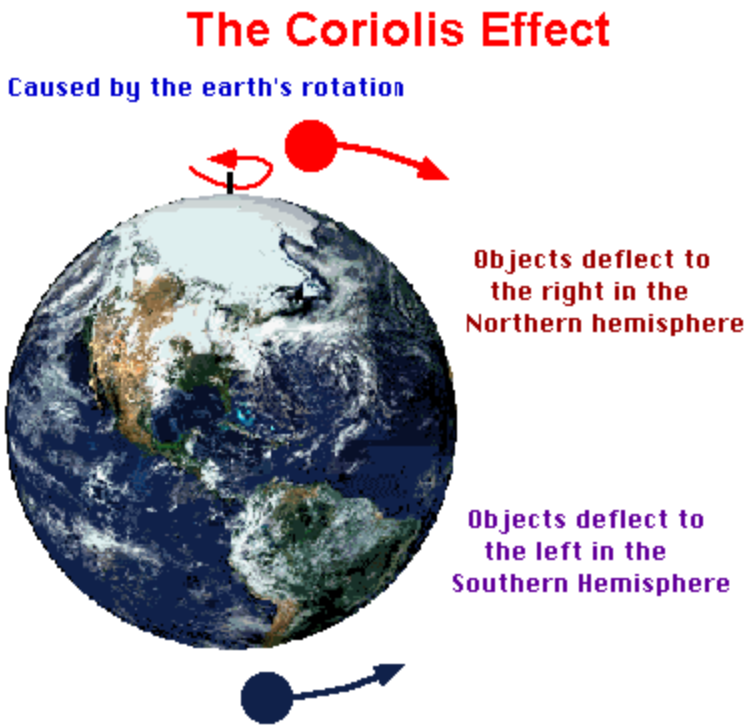


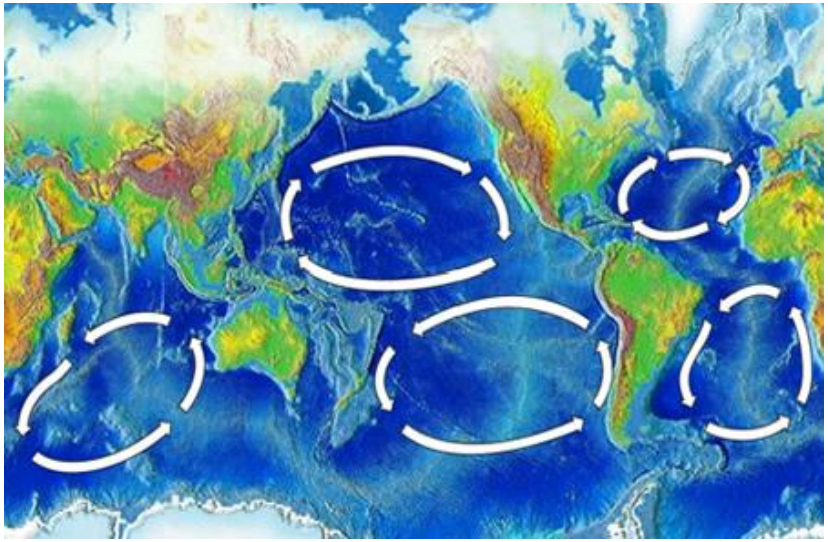
Continental Barriers

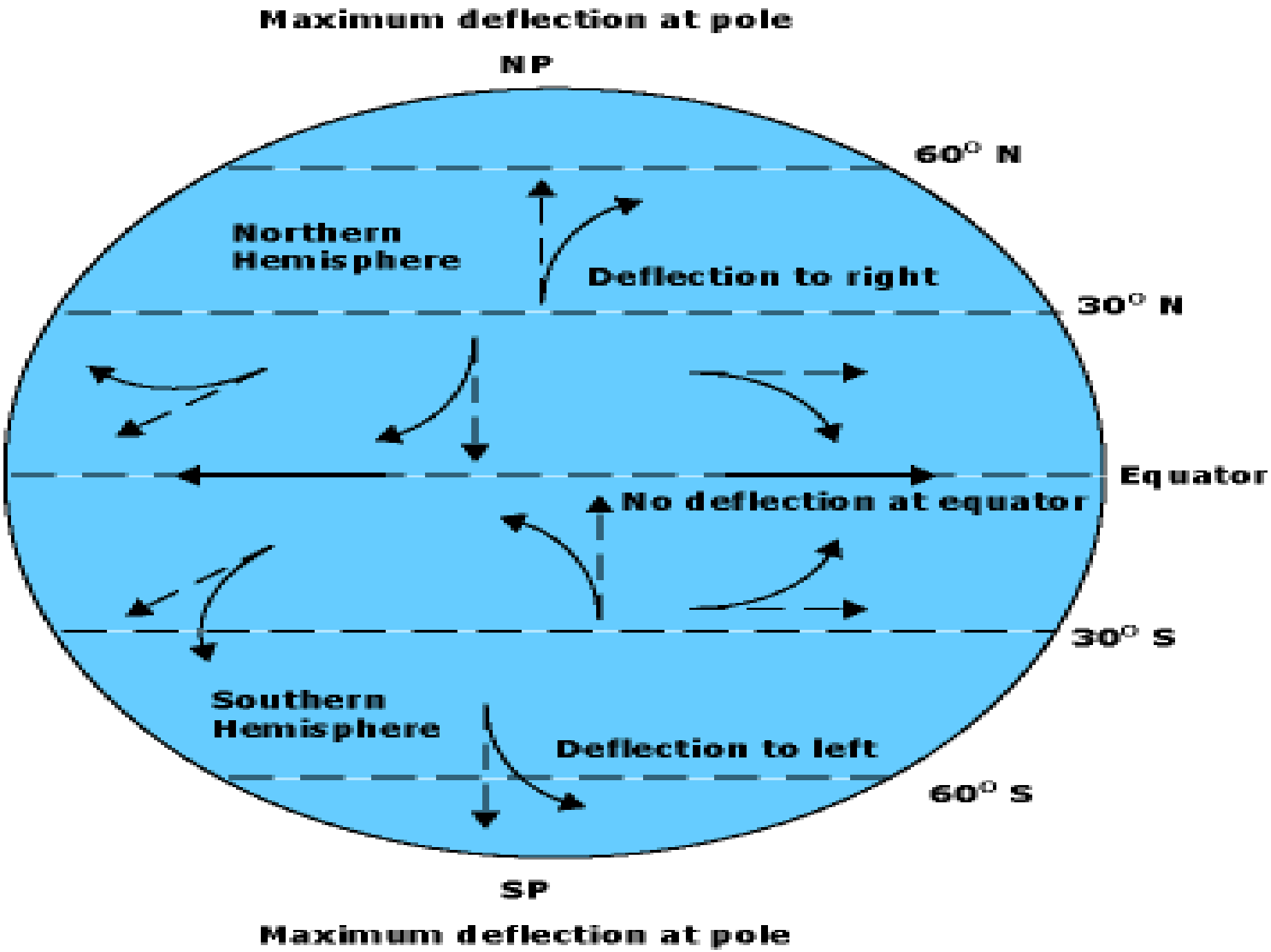
- The continents are another major influence on surface currents.
  - They act as barriers to these currents.
  - When a surface current flows against a continent, the current is deflected and divided.

Coriolis Effect

- As Earth rotates, ocean currents and wind belts curve.
  - The curving of the paths of ocean currents and winds due to Earth's rotation is called the Coriolis Effect.
  - The wind belts and the Coriolis Effect create huge circles of moving water, called gyres.

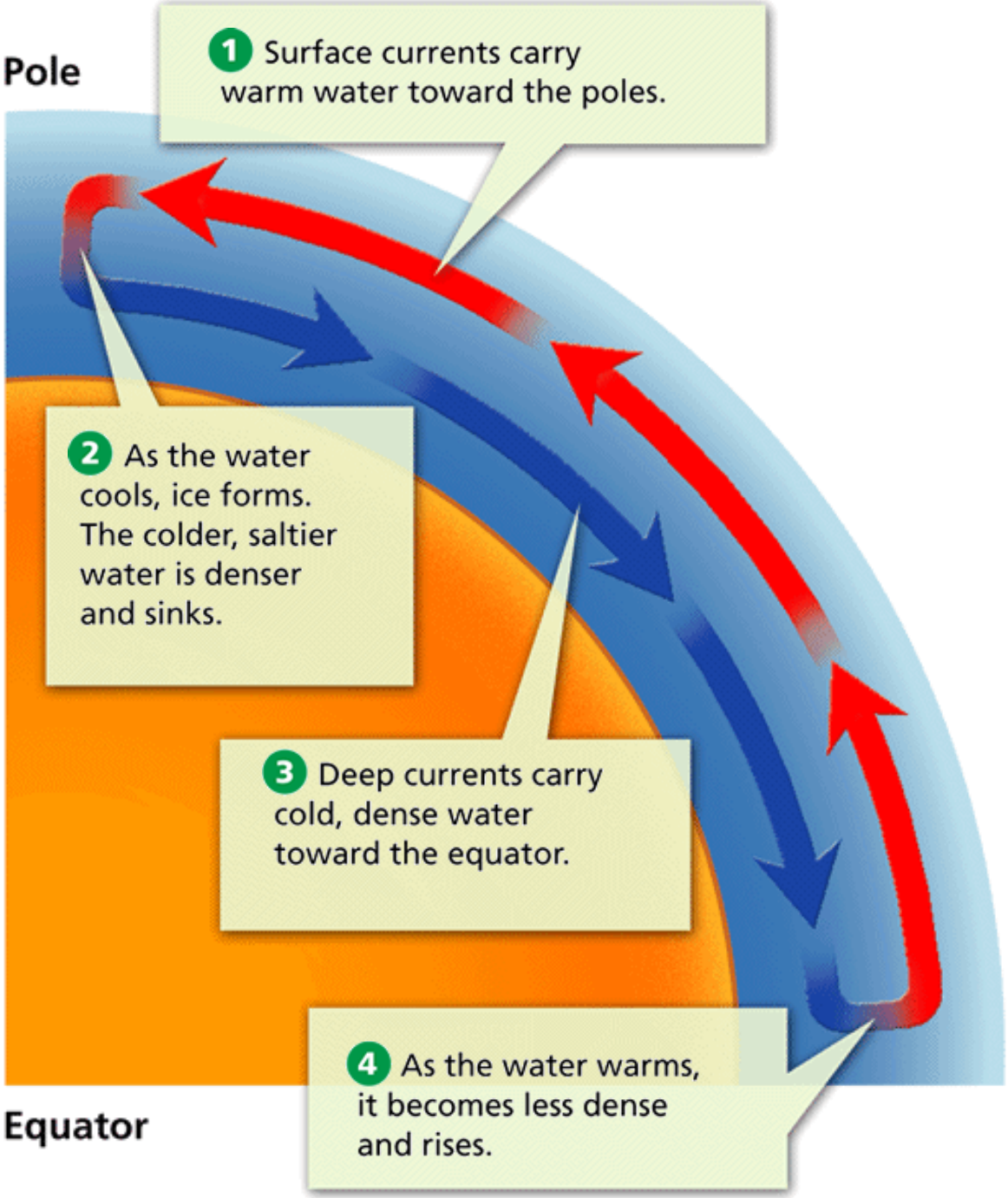




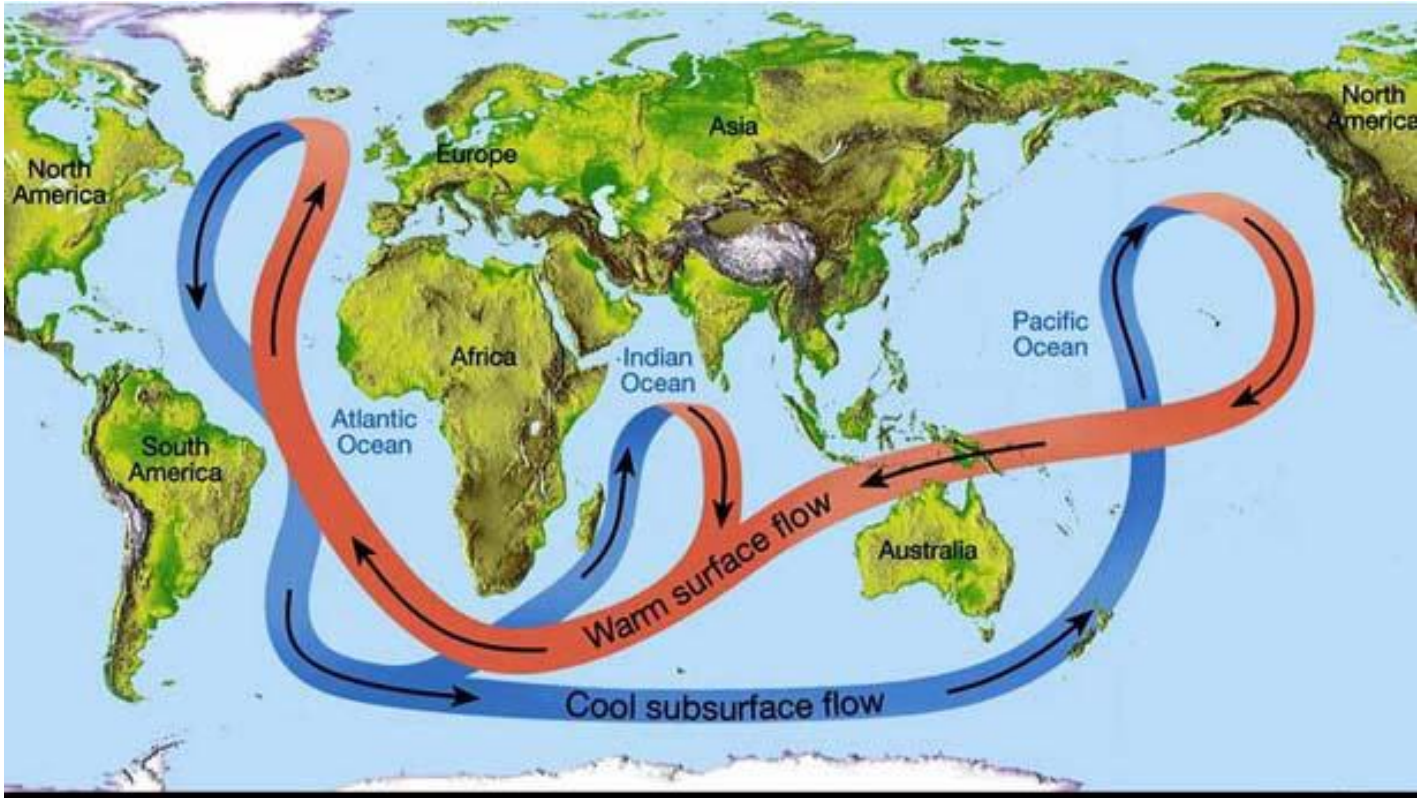

Maximum deflection at pole
NP
60° N
Northern Hemisphere
Deflection to right
30° N
Equator
No deflection at equator
30° S
Southern Hemisphere
Deflection to left
60° S
SP
Maximum deflection at pole

Deep Currents

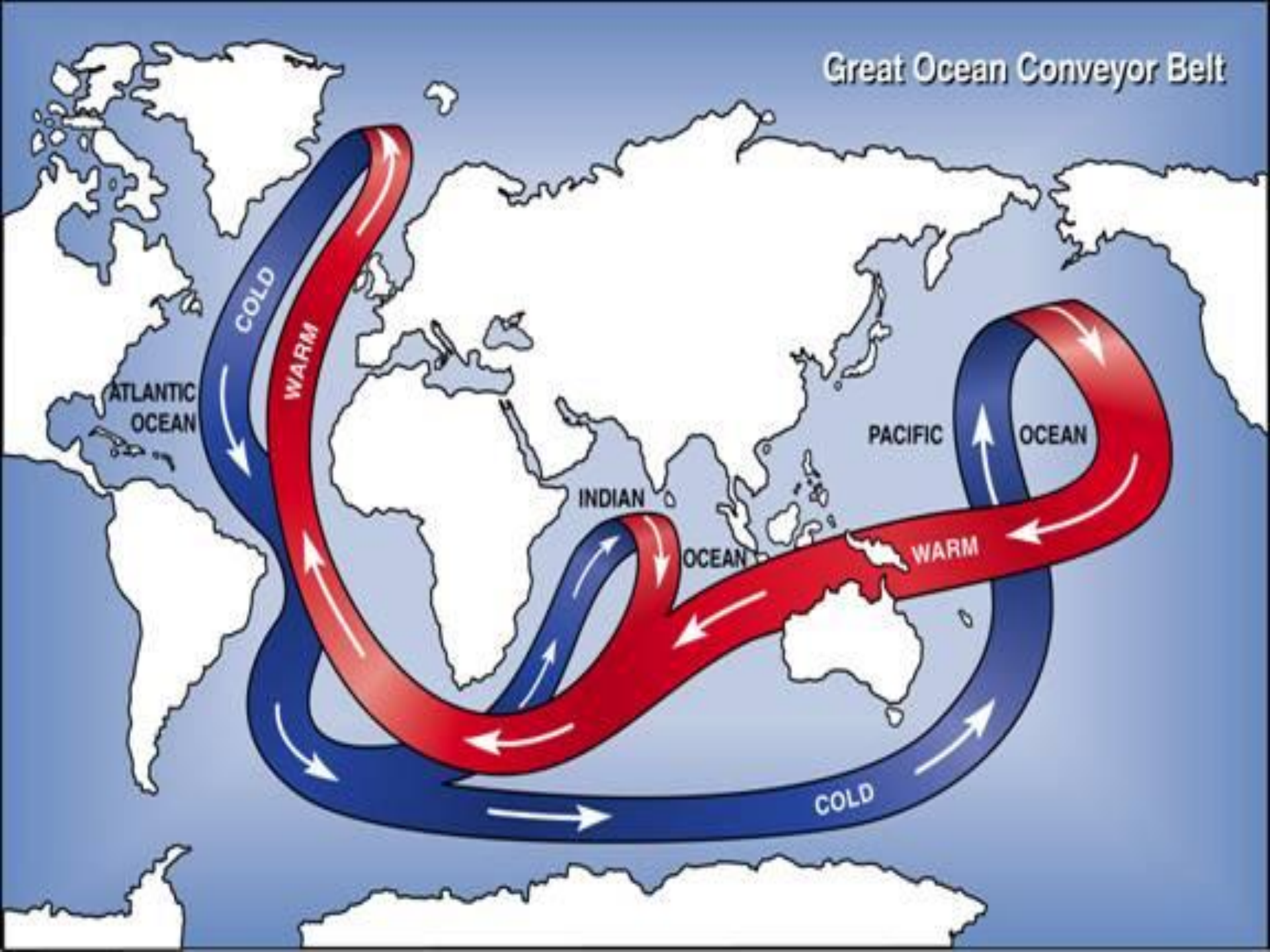
a. Stream-like movements of ocean water located far below the surface are called deep currents.
b. Move much *slower* than surface currents.
c. Form as cold, dense water of the polar regions sinks and flows beneath warmer ocean water.
    - The density of ocean water if affected by temperature and salinity.
        - Decreasing temperature and increasing salinity will increase the water's density.
        - Cold water is more dense than warm water!


Pole
1 Surface currents carry warm water toward the poles.
2 As the water cools, ice forms. The colder, saltier water is denser and sinks.
3 Deep currents carry cold, dense water toward the equator.
4 As the water warms, it becomes less dense and rises.
Equator

d. when combined with surface currents, results in conveyor belt movement of water around globe

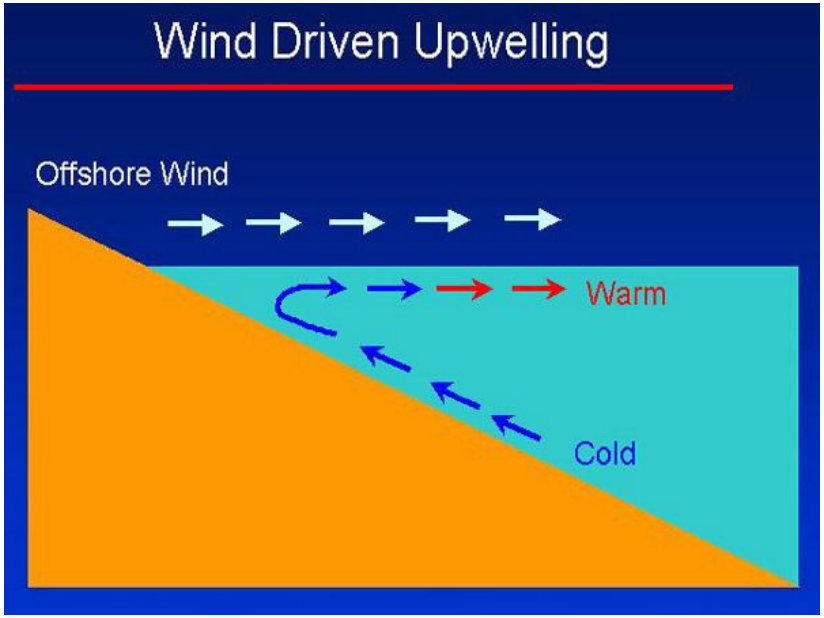


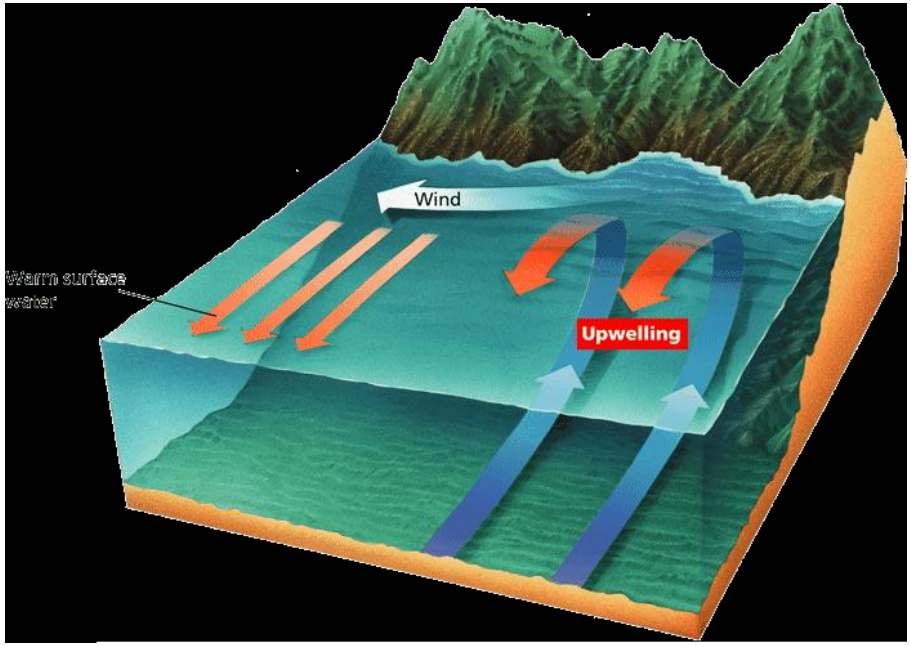
THERMOHALINE CIRCULATION - GREAT OCEAN CURRENT


Great Ocean Conveyor Belt
COLD
WARM
ATLANTIC
OCEAN
INDIAN
OCEAN
PACIFIC
OCEAN
WARM
COLD

upwelling

a. wind blows, moves water away, causes new water to rise up to replace it
b. brings up tiny ocean organisms, minerals, and other nutrients from the deeper layers of the water.

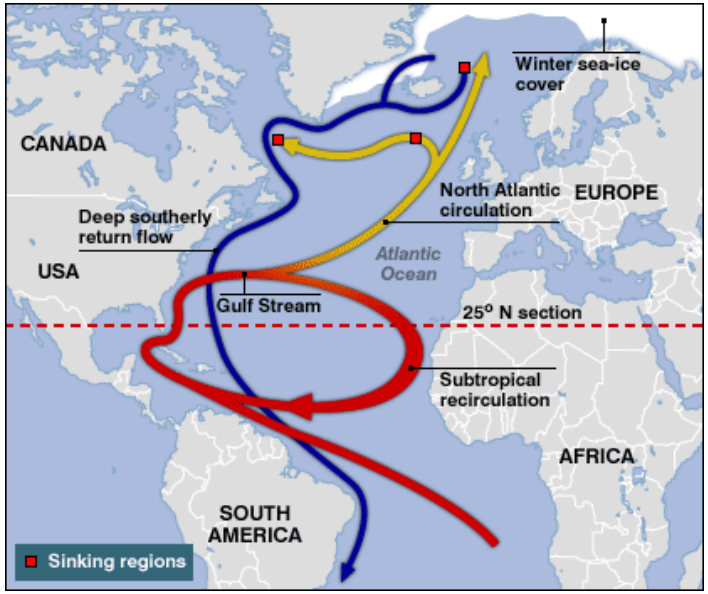




Currents and Climate

1. Currents can greatly affect the climate in many parts of the world.
   - Warm-water currents:
     - The Gulf Stream carries warm water from the Tropics to the North Atlantic Ocean.
   - Cold-water currents:

     For example: The California current carries cold water from the North Pacific Ocean toward Mexico along the western coast of the USA → therefore, cooler climate year-round than inland states.



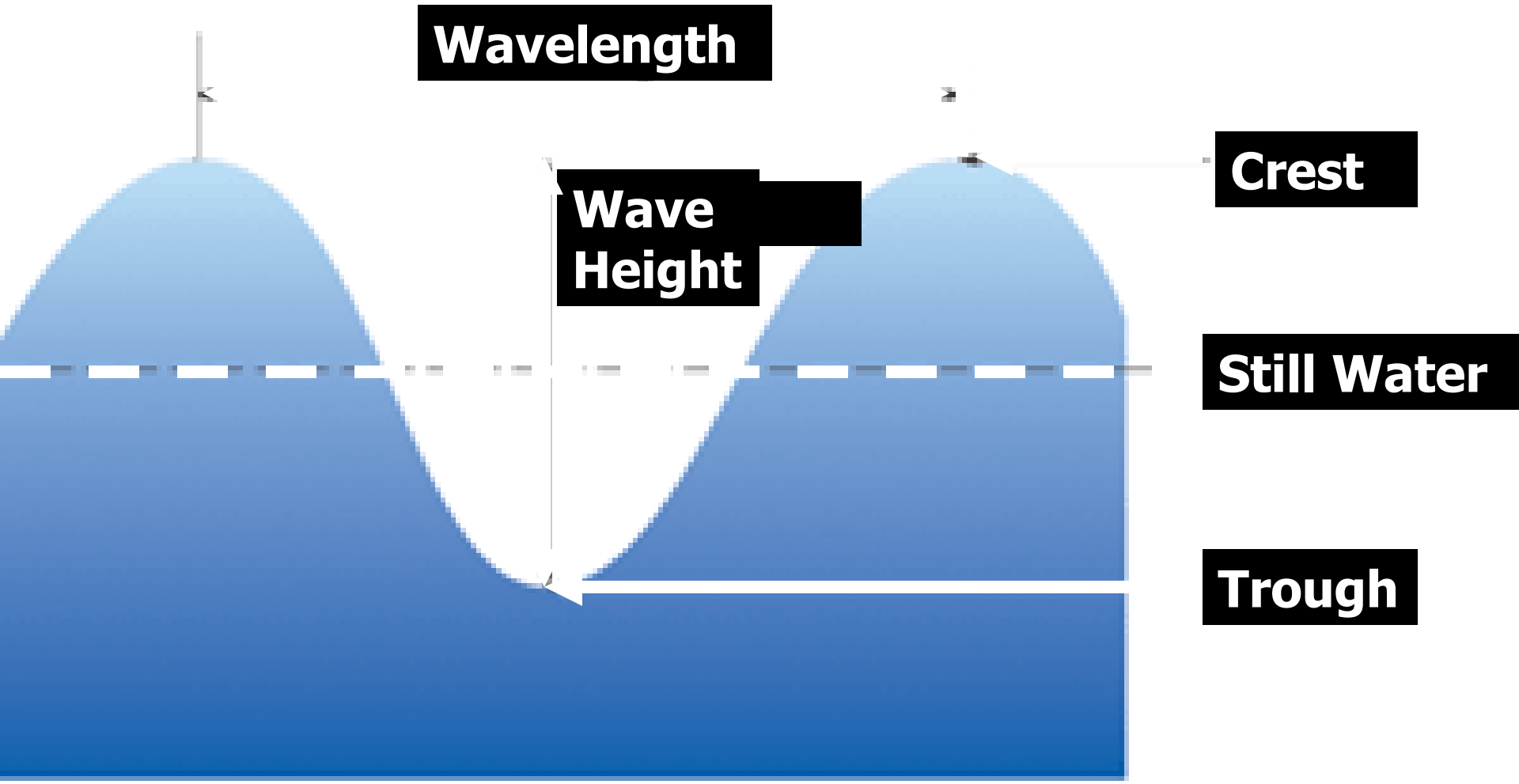
Parts of a Wave

a. **Crest – highest point of a wave**

b. **Trough – lowest point of a wave**

c. **Wave Height – vertical distance between the crest and the trough**

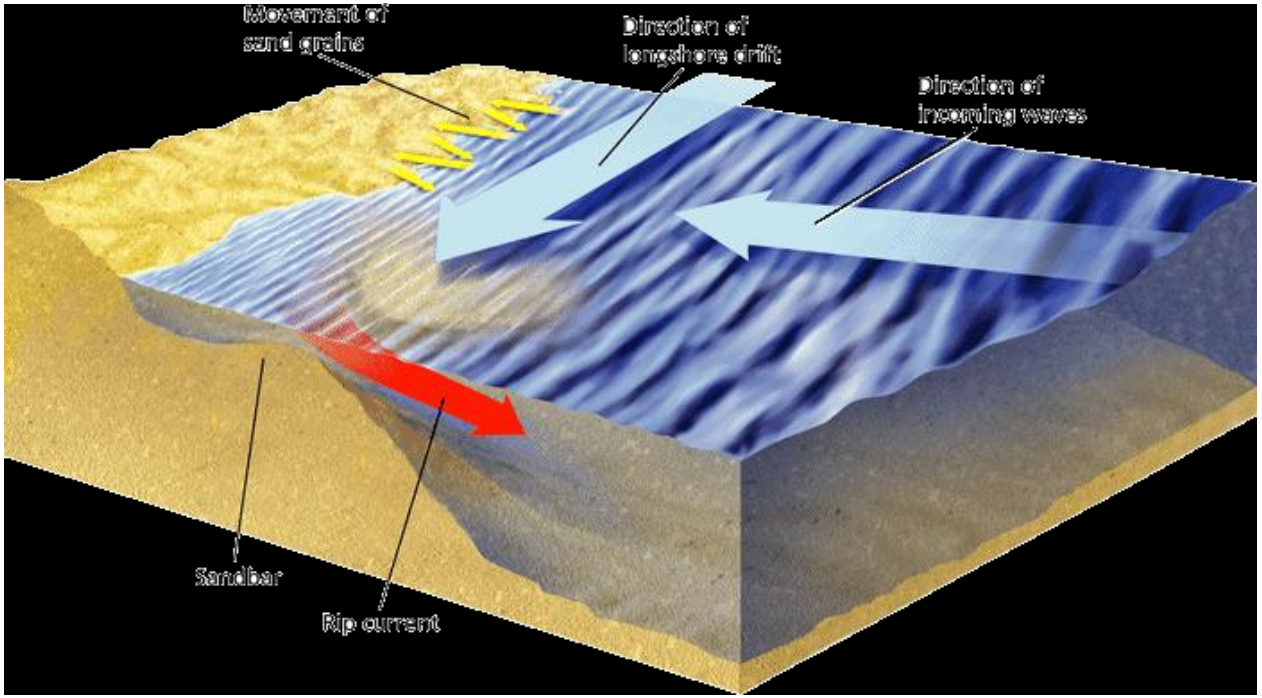
d. **Wavelength – horizontal distance between two crests or two troughs**



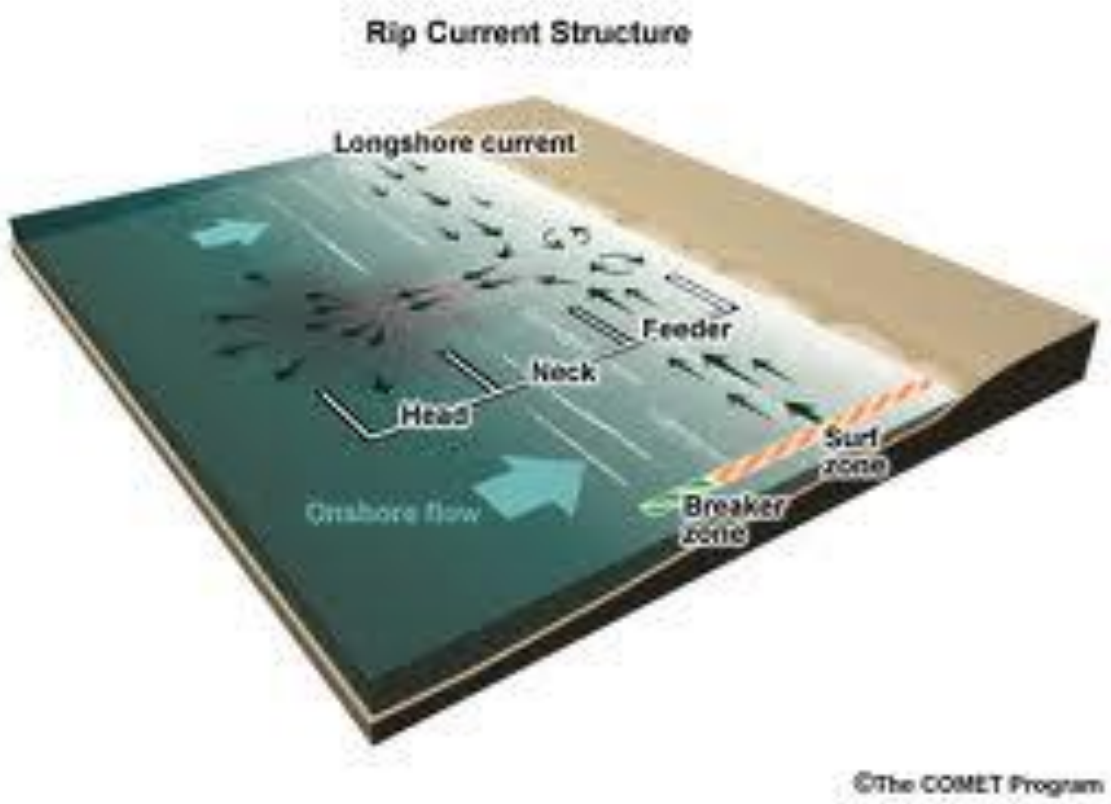
Wave Parts

Effects of Waves on Shore

a. Long shore current- As waves come into shore, water washes up the beach at an angle, carrying sand grains. The water and sand then run straight back down the beach.



Rip Current-long ridges or piles of sand create sand bars. A break in a sand bar allows a fast-moving narrow stream of water through



Cause of Waves

a. Wind

- When wind blows across a body of water, friction causes the water to move along with the wind.
- Wave Height depends on –
  - Wind speed
  - Distance over which the wind blows
  - Length of time the wind blows

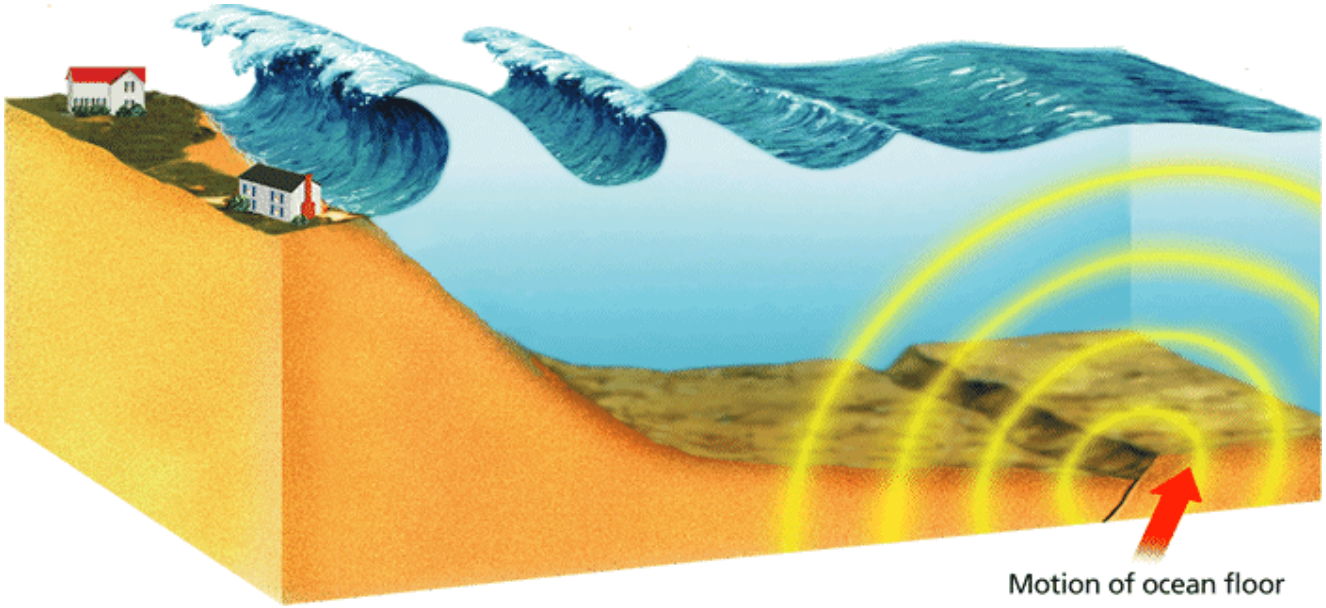
Earthquakes- Waves caused by earthquakes are called Tsunamis

i. Tsunamis were once called Tidal waves, but they have nothing to do with the tides.

ii. They are produced by earthquakes and other seismic disturbances. That's why they're also called <u>seismic sea waves.</u>

iii. Sudden outflow of water then it returns much higher

iii. Tsunamis are very long, fast moving waves!



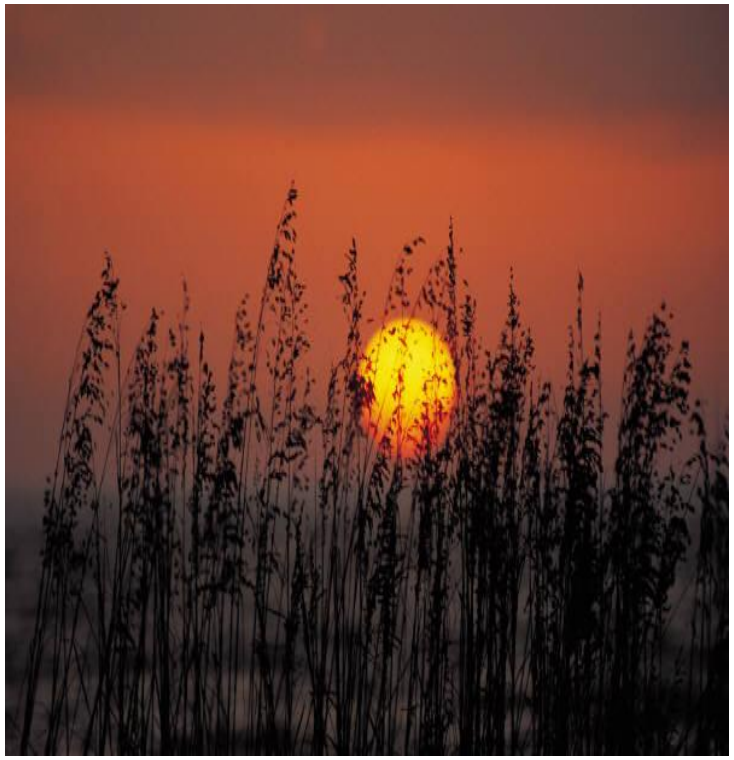
- They can have wavelengths of 150 miles, wave heights of 100 ft and move as fast as 450 mph (jet speed!).

Gravitational force of the Moon and Sun creates Tides (pgs 96-100)..

i. The rise and fall in sea level is called a tide.

ii. One low-tide/high-tide cycle takes about 12 hrs and 25 min.

iii. Tidal range is the difference in ocean level between high-tide and low-tide



What is the Tidal Range?

- HT = 30 ft, LT = 20 ft **10 ft**
- HT = 20 ft, LT = 12 ft **8 ft**
- HT = 50 ft, LT = 20 ft **30 ft**

Gravitational Effect of the Moon

- Two big bulges of water form on the Earth:
  - one directly under the moon
  - another on the exact opposite side
- As the Earth spins, the bulges follow the moon.
- These are normal daily tides

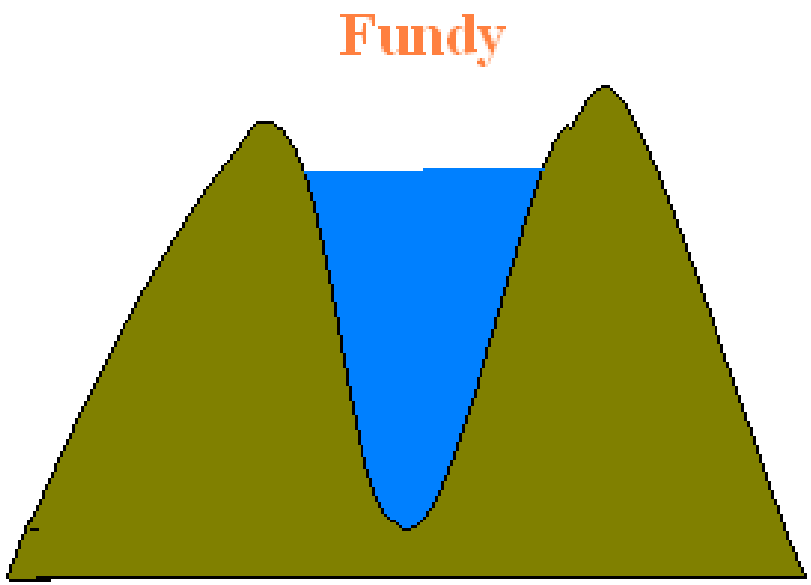
Barrage Logistics

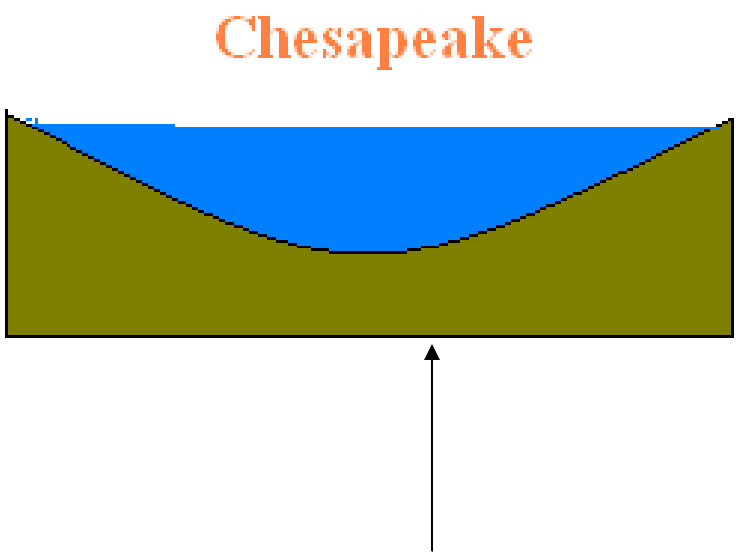
- Largest surface area, greatest tidal range
  - Square Effect
- Effects of ecosystem (fish, sediment, etc.)
- Proximity to population

Waterway Measurement Logistics

	Mean Tidal Amplitude(m)	**Basin Area (km^2)**
La Rance, France	4	17
Bay of Fundy, Canada	5.5	240
Annapolis, Nova Scotia	3.2	6
Severn Estuary	4	420
Garolim Bay, South Korea	2.5	85

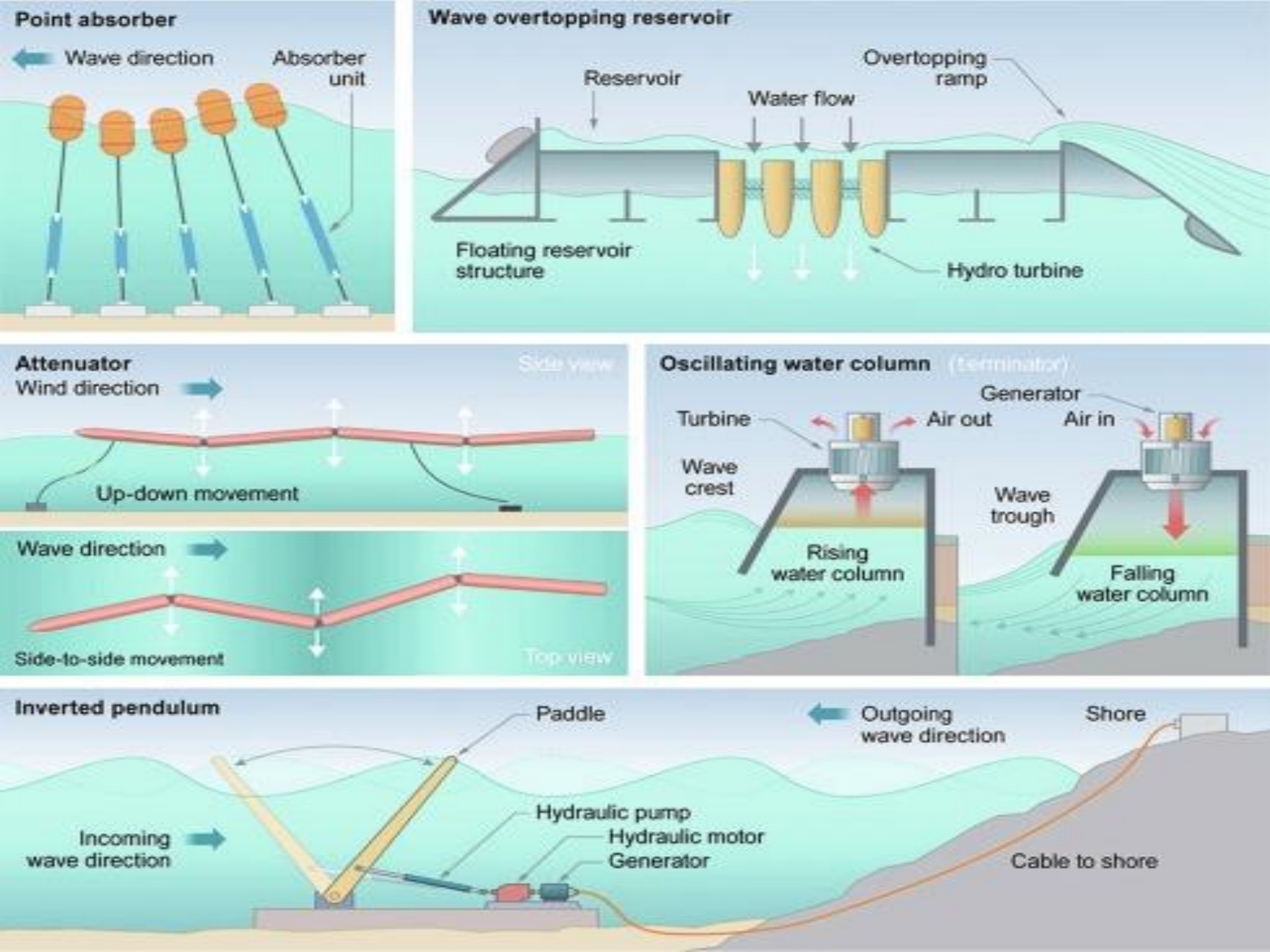
Bay of Fundy VS Chesapeake Bay





Types of WECs

- **Surface Devices:** Float on the surface and rise and fall with the waves, driving turbines or other mechanisms.
- **Submerged Devices:** Operate below the surface, using the wave motion to drive turbines or pumps.
- **Overtopping Devices:** Use waves to fill a reservoir, which is then drained through a turbine.
- **Oscillating Water Columns:** Use the rise and fall of water in a chamber to compress air, which then drives a turbine.


Point absorber
Wave direction
Absorber unit
Wave overtopping reservoir
Reservoir
Overtopping ramp
Water flow
Floating reservoir structure
Hydro turbine
Attenuator
Side view
Wind direction
Up-down movement
Wave direction
Side-to-side movement
Top view
Oscillating water column
(terminator)
Generator
Turbine
Air out
Air in
Wave crest
Wave trough
Rising water column
Falling water column
Inverted pendulum
Paddle
Outgoing wave direction
Shore
Hydraulic pump
Hydraulic motor
Generator
Incoming wave direction
Cable to shore

Attenuators:

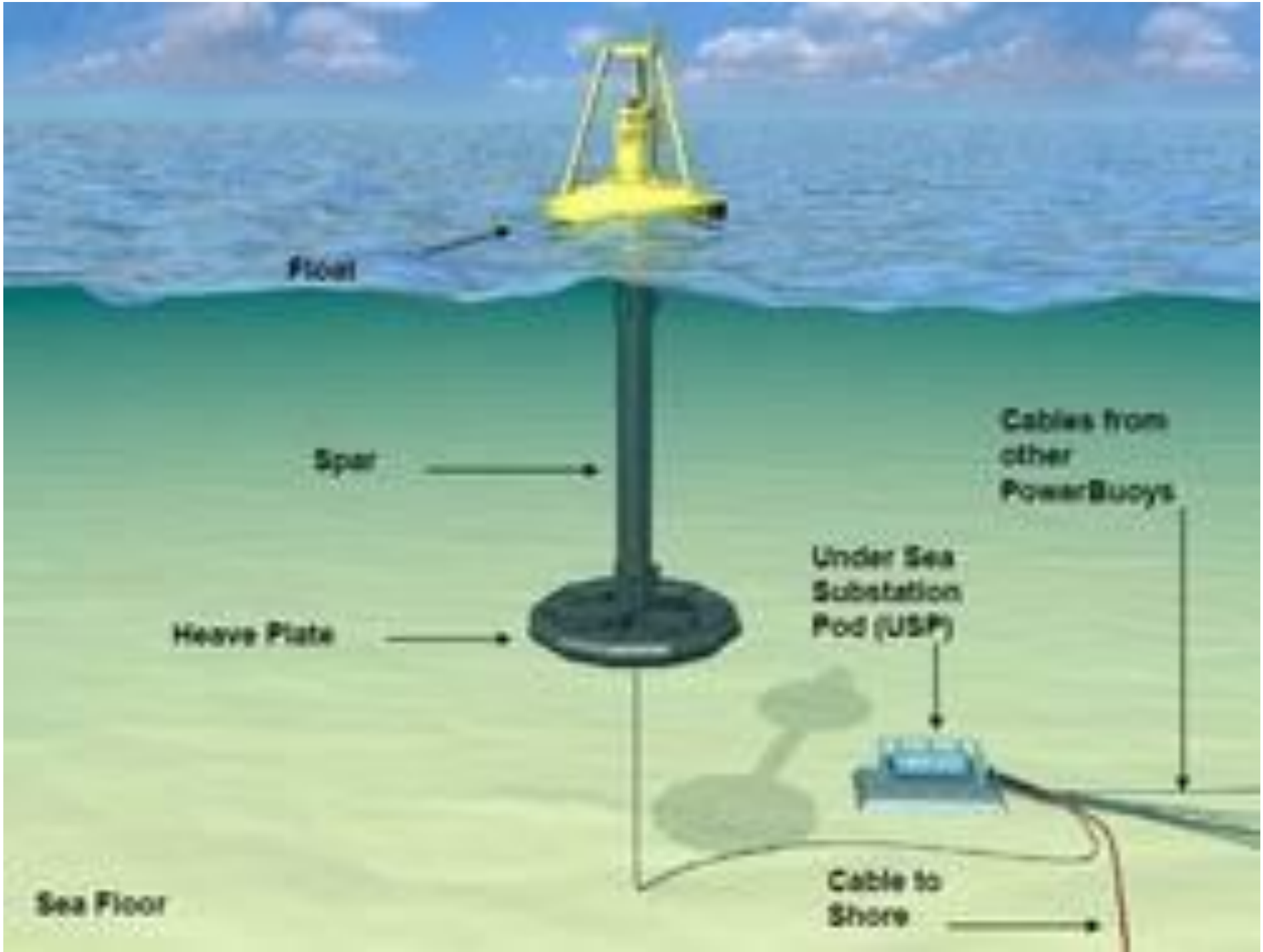
- These WECs are designed to reduce the height of incoming waves, thereby extracting energy as the waves pass through.

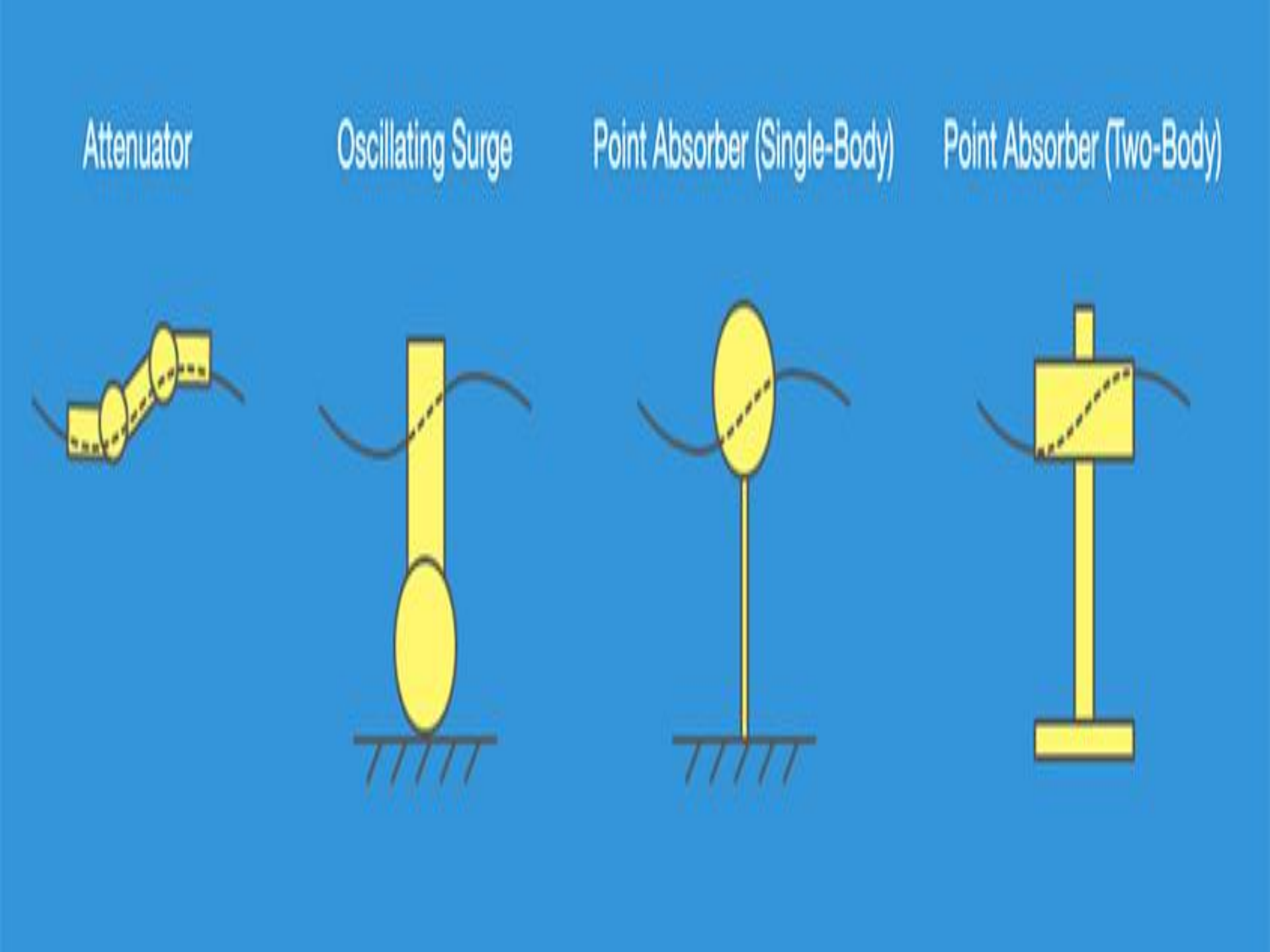
Point Absorbers:

- These systems consist of a buoy that moves with the waves, converting the wave energy into mechanical energy, which is then used to generate electricity.

Oscillating Wave Surge Converters:

- These systems utilize the pressure fluctuations of waves to drive a turbine or piston, converting wave energy into mechanical energy.


Float
Spar
Heave Plate
Cables from other PowerBuoys
Under Sea Substation Pod (USP)
Cable to Shore
Sea Floor


Attenuator
Oscillating Surge
Point Absorber (Single-Body)
Point Absorber (Two-Body)

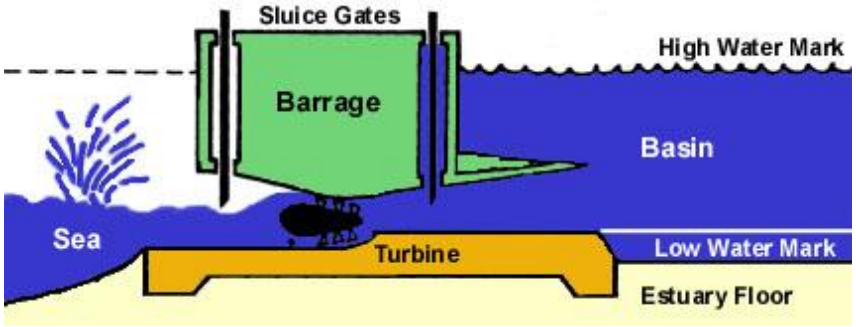
New Technology from an Old Idea

Industrial Revolution

- Brought cheap energy in the form of electricity
- Small tidal mills were replaced with large steam-powered roller mills
- Paleotechnic period means full steam ahead with little though of what was happening to the environment

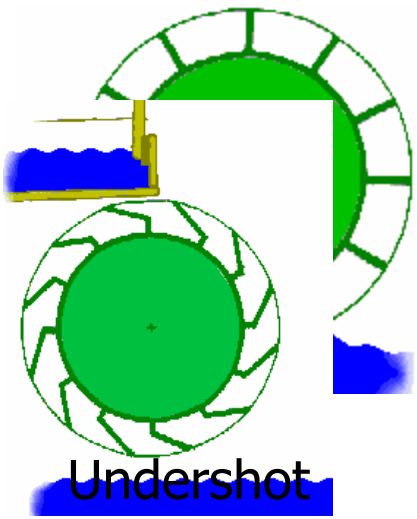
From Milling to Electricity

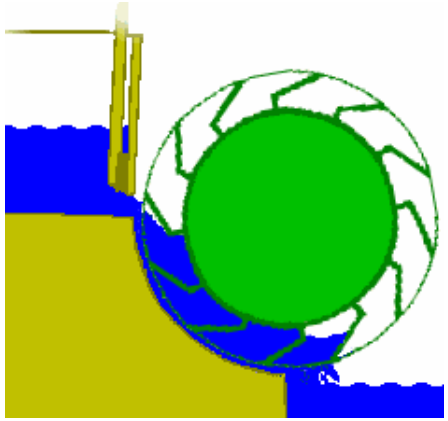
- Most common generating system is the ebb generating system
- Double effect turbines are now becoming technologically feasible



Evolution of Turbine Types

- Waterwheel turbines were used in tidal mills



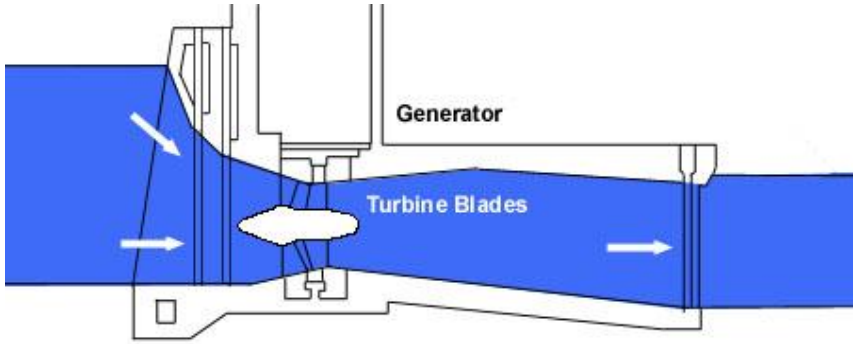


Breast-shot

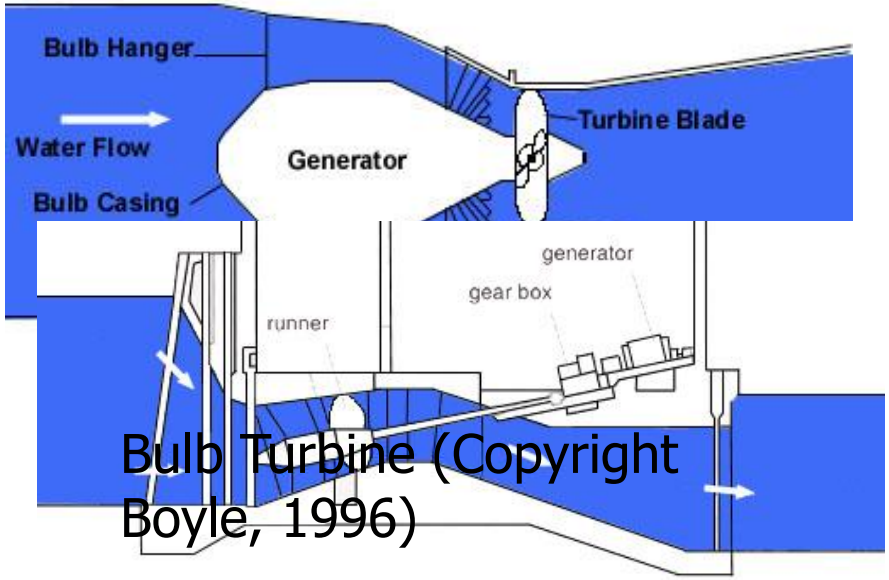
Overshot

Recent Turbines

- Bulb, Rim, and Tubular



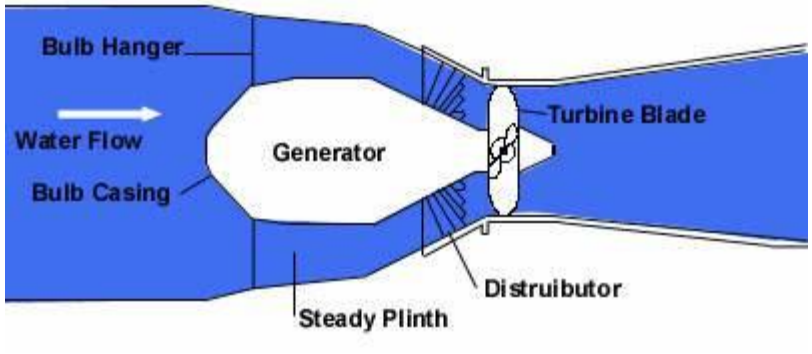
Rim Turbine (Copyright Boyle, 1996)



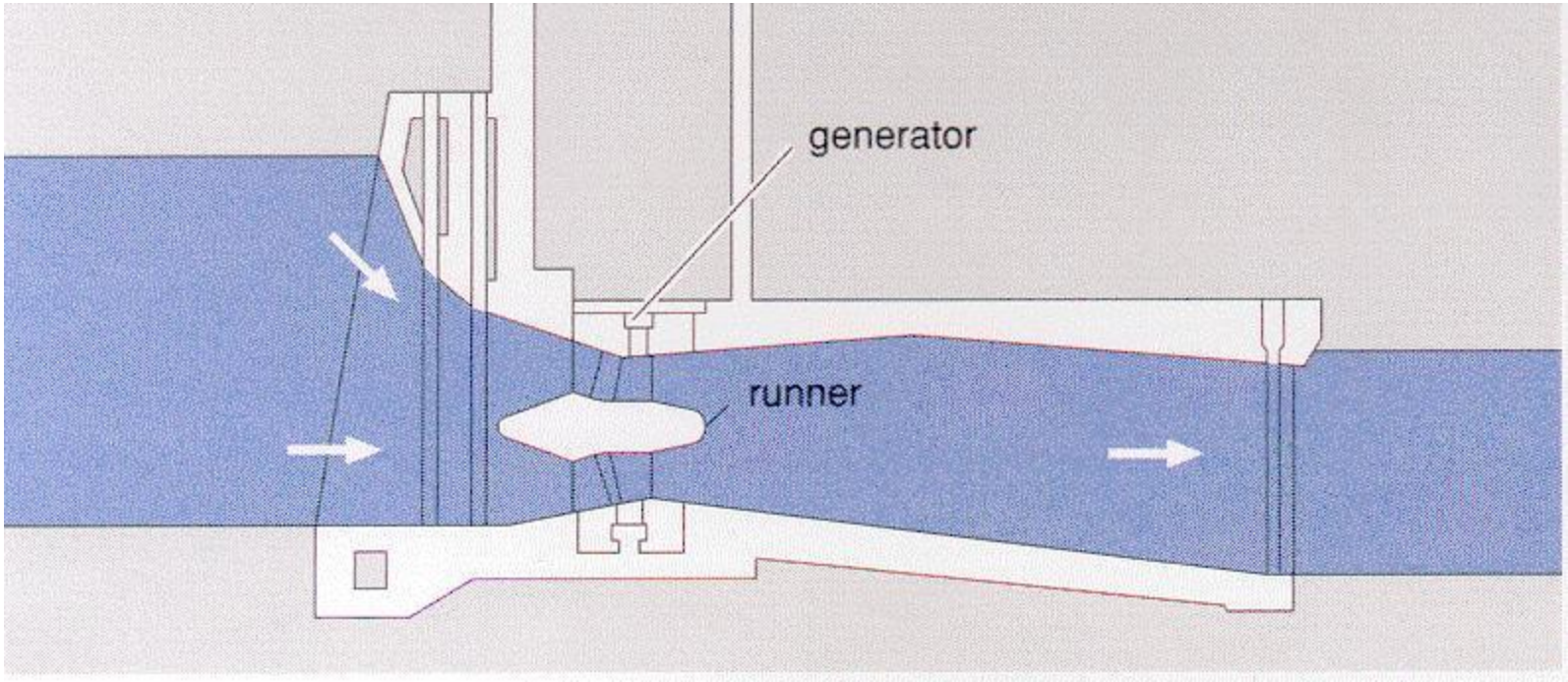
Bulb Turbine (Copyright Boyle, 1996)

Tubular Turbine (Copyright Boyle, 1996)

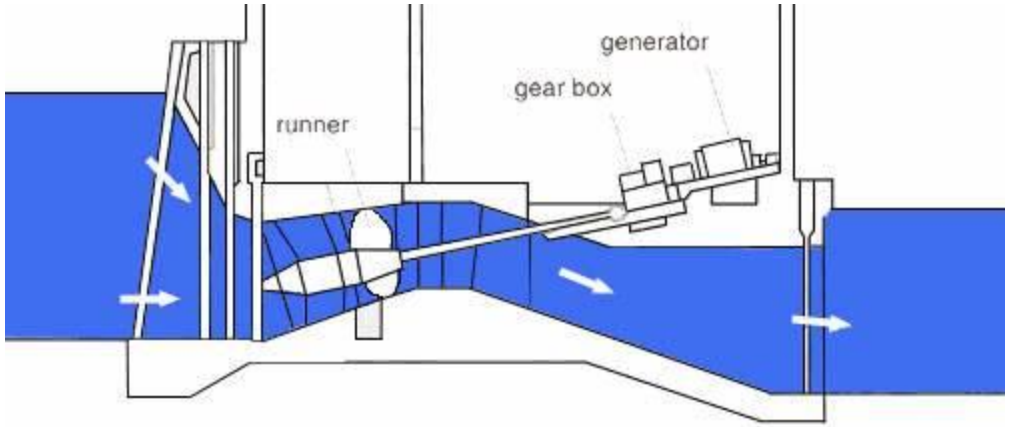
Bulb Turbine



Straflo or Rim Generator Turbine

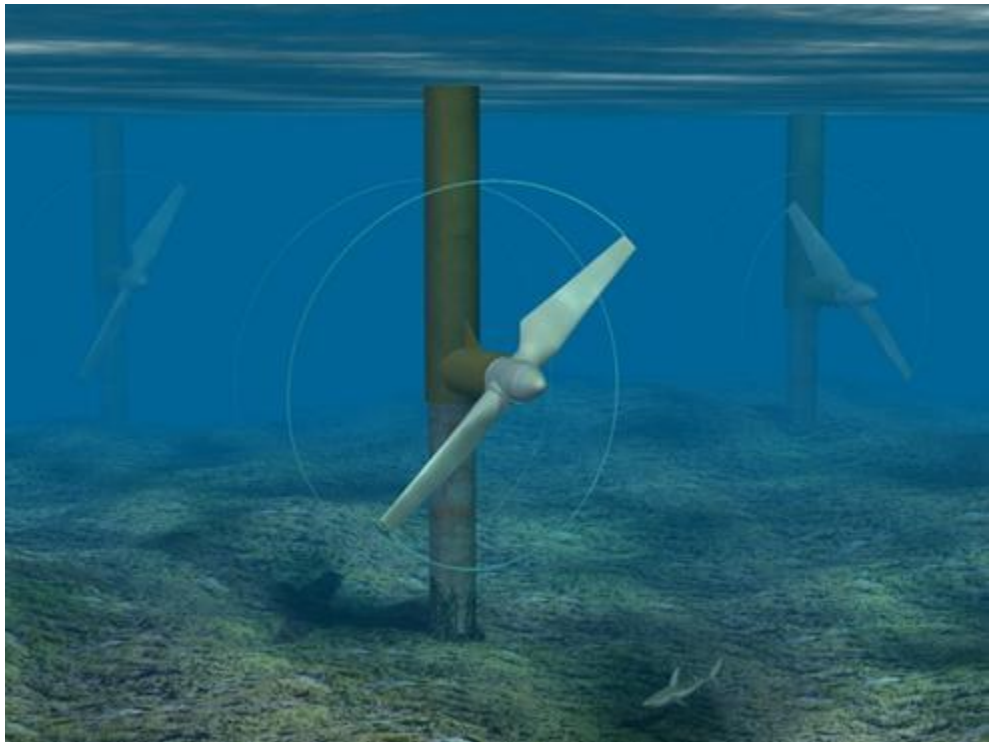


Tubular Turbine



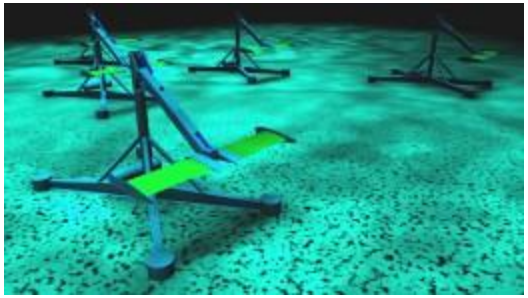
Tidal streams

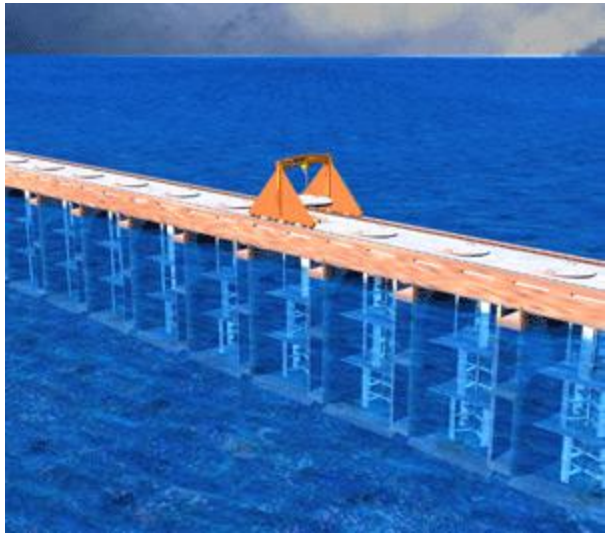
- Instead of damming estuaries the tidal currents are harnessed using wind like turbines



Main methods of current generation

- Marine current generator
- Stingray
- Tidal fence





Types of Tidal Power Generation

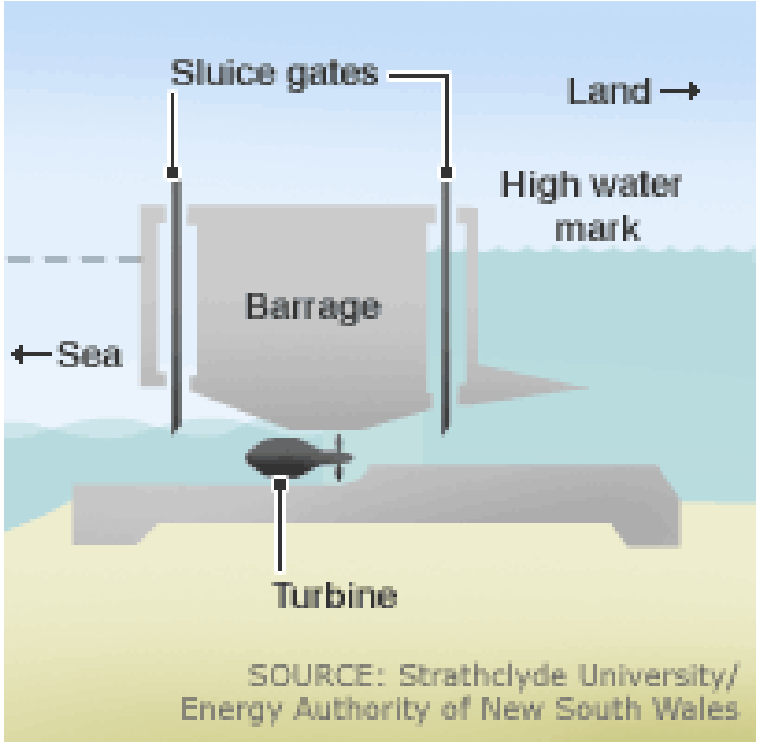
- Barrages
- Tidal Streams

Tidal Barrage

Similar to a dam

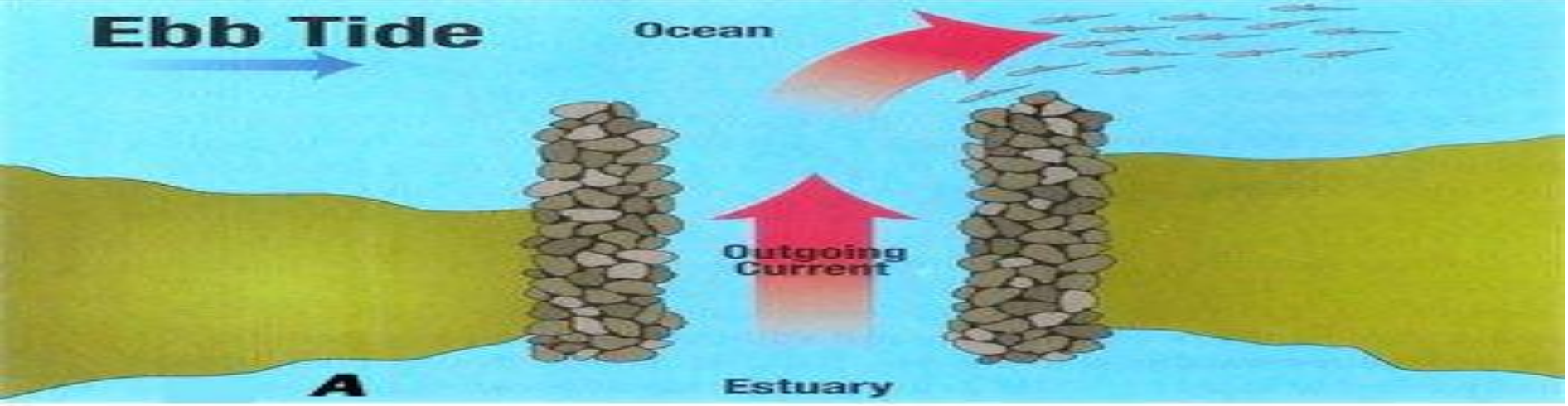
Structure must span the entire width and height of the estuary

- Turbines
- Sluice gates
- Embankments



Ebb and Flood Currents

- A horizontal movement of water often accompanies the rising and falling of the tide: tidal current
- Incoming tides = flood current
- Outgoing tides = ebb current
- Strongest before or near the time of high/low tide
- Weakest between high/low tide: slack tides

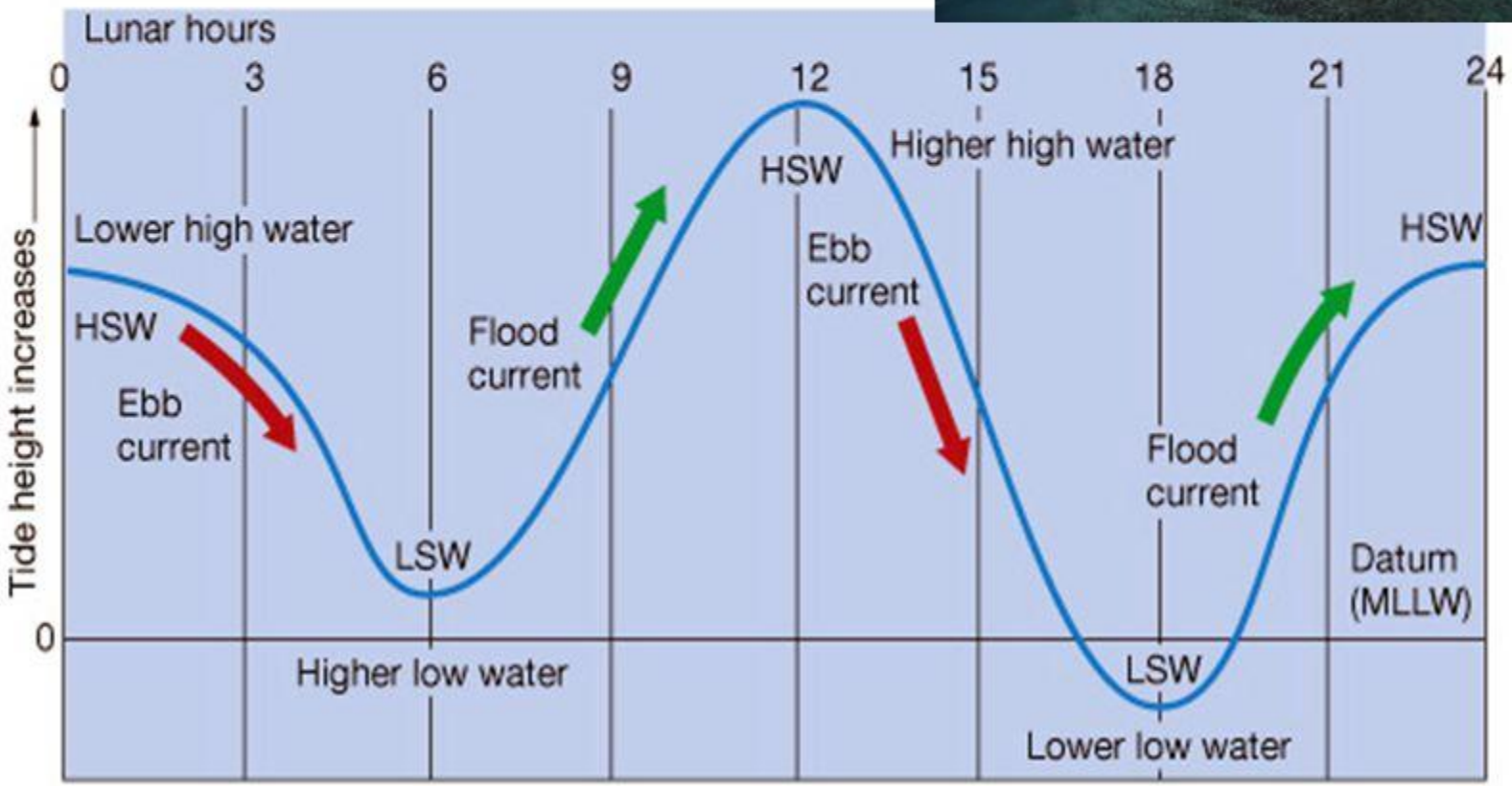

Ebb Tide
Ocean
Outgoing Current
A
Estuary

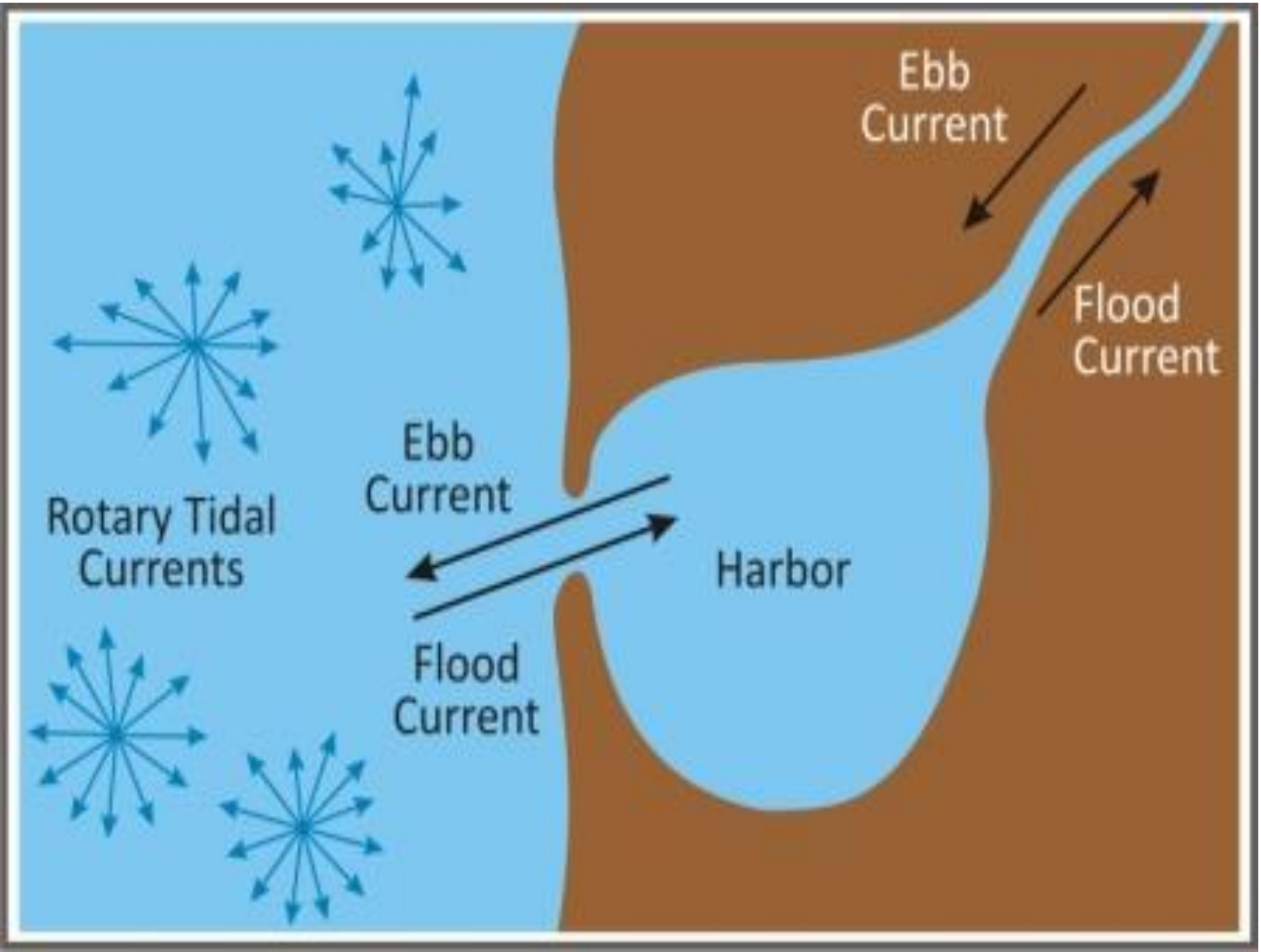
Slack Tide
Ocean
Outgoing Current
B
Estuary

Ocean
Flood Tide
Outgoing Current

Tidal Currents

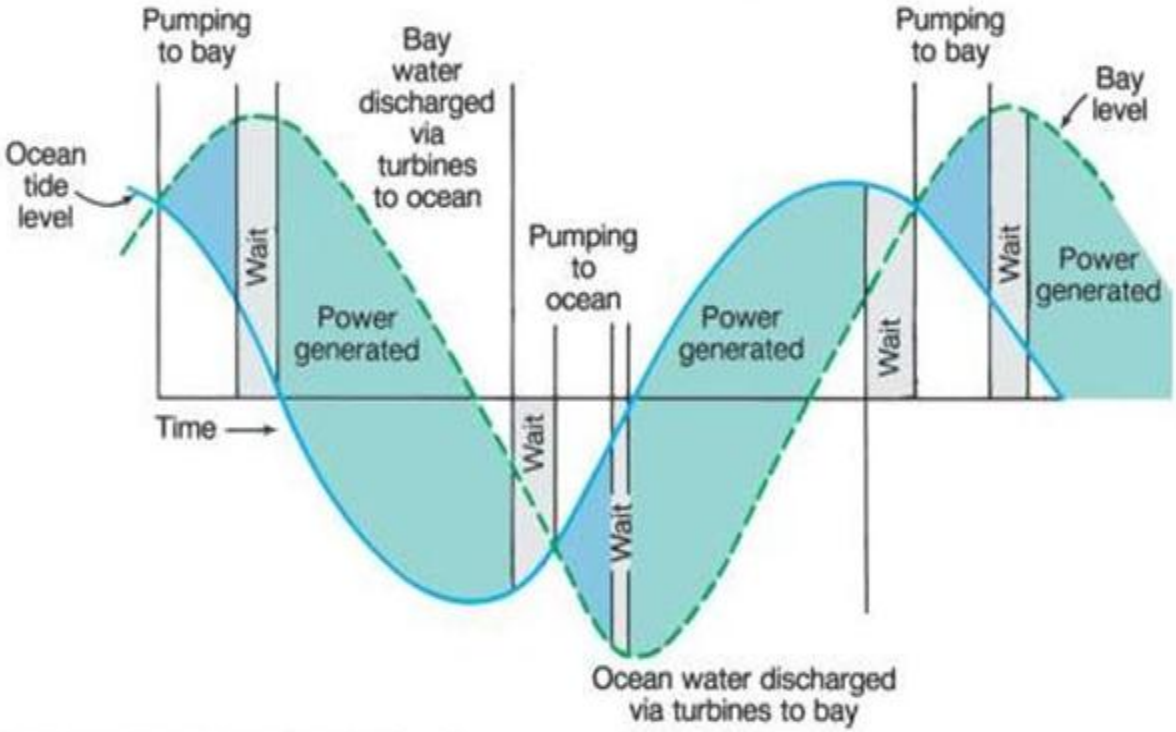
Tidal currents occur in some bays and rivers due to a change in tides



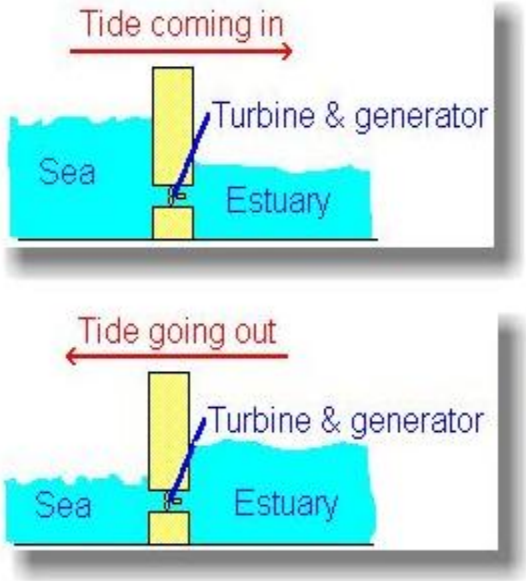

Ebb
Current
Flood
Current
Ebb
Current
Rotary Tidal
Currents
Harbor
Flood
Current

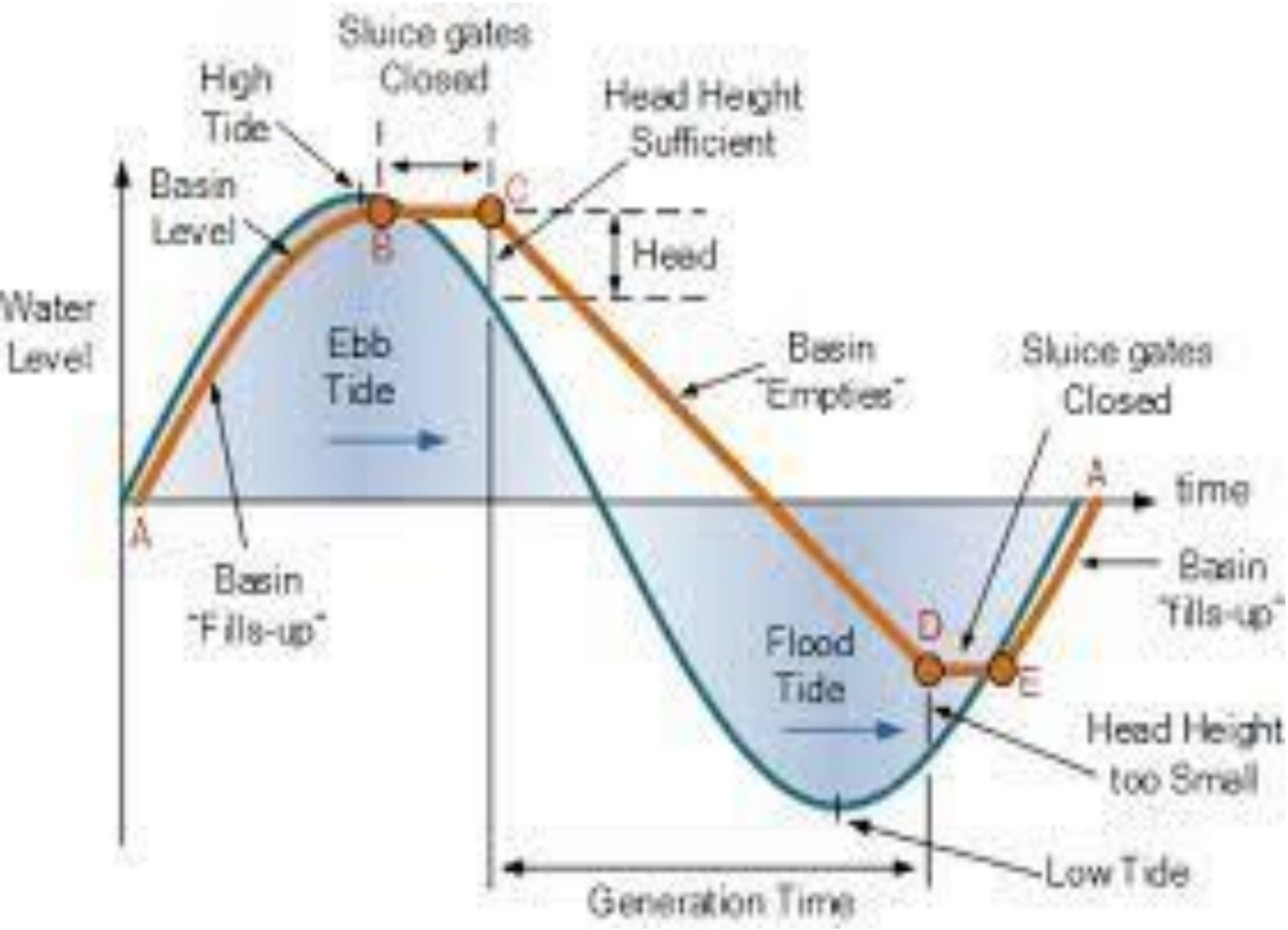
Tidal Energy

- Two systems to extract energy from tides:
  - Double-action power cycle



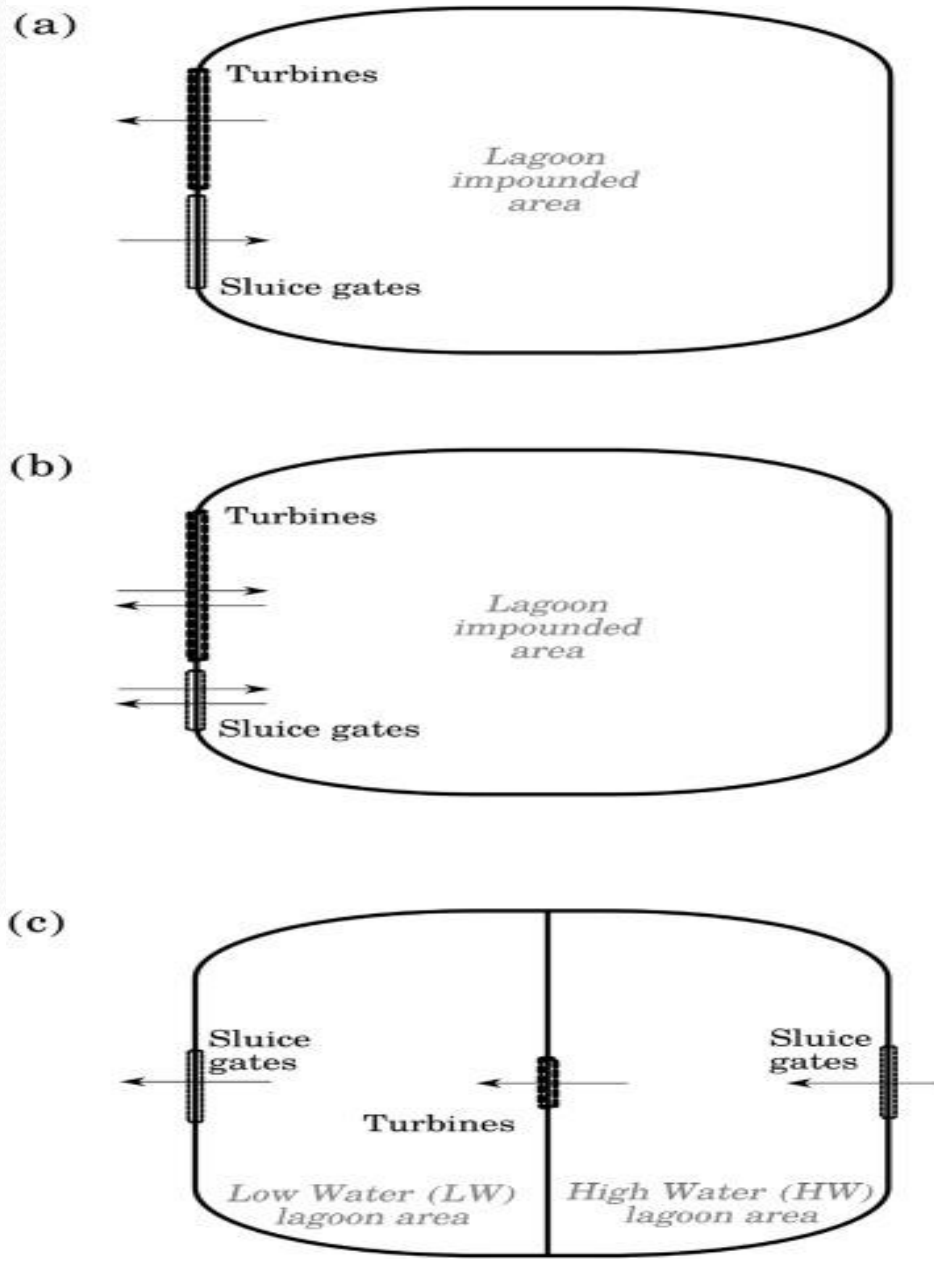
(b) Double-action power cycle; ebb and flood



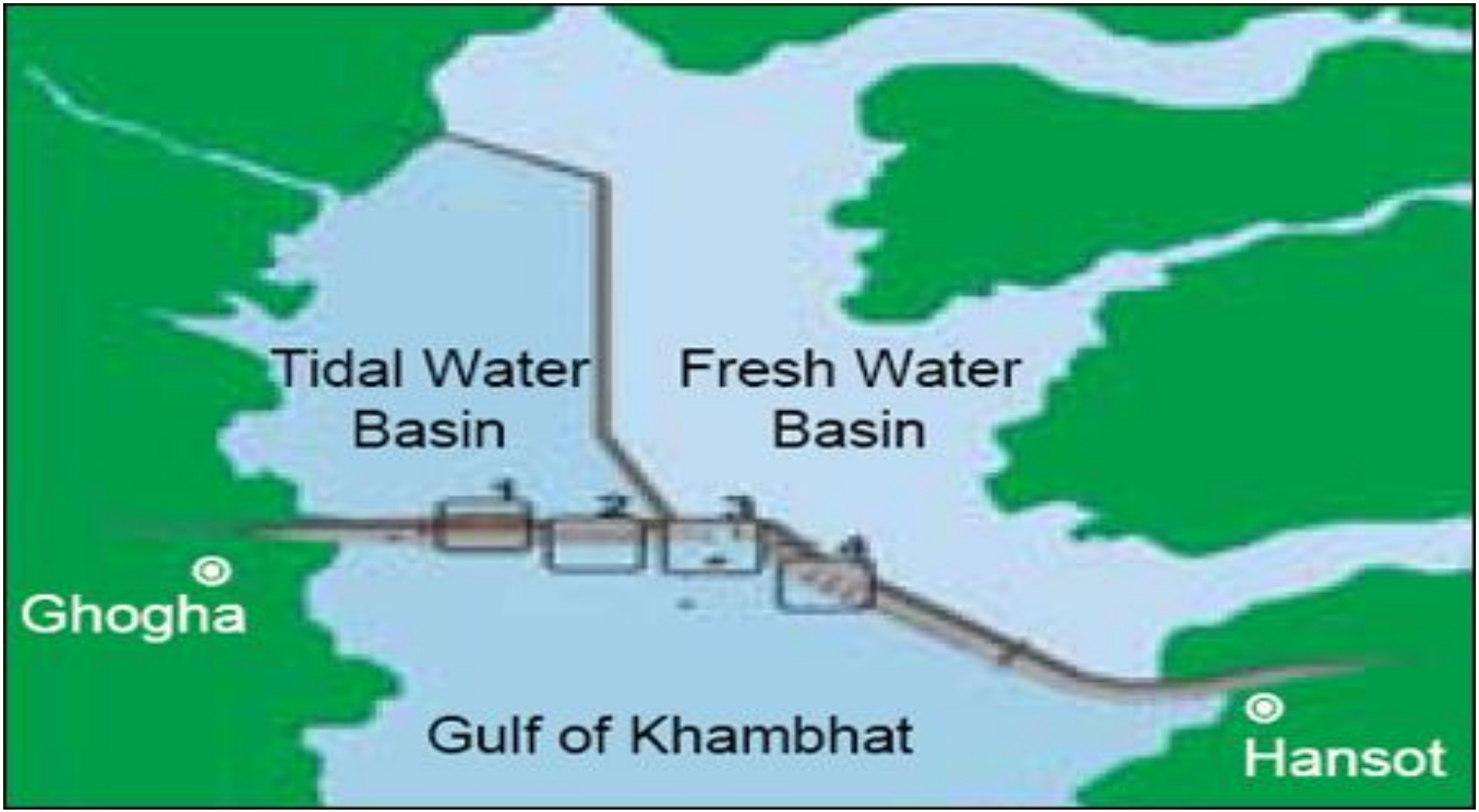

Sluice gates
Closed
High
Tide
Head Height
Sufficient
Basin
Level
C
B
Head
Water
Level
Ebb
Tide
Basin
"Empties"
Sluice gates
Closed
A
time
A
Basin
"Fills-up"
Basin
"fills-up"
D
Flood
Tide
E
Head Height
too Small
Low Tide
Generation Time

Types of tidal power plants

- Kinetic energy
  - Tidal fences
  - Tidal farms
- Potential energy
  - Barriage type/ basin type
    - Flood generation
    - Ebb generation
    - Two way generation



Gulf of Khambhat



Basic Concepts of Tidal Energy in India

- https://www.toppr.com/bytes/tidal-energy-in-india/

- https://www.greenfacts.org/en/tidal-energy/l-2/index.htm
- https://en.wikipedia.org/wiki/Tidal_barrage

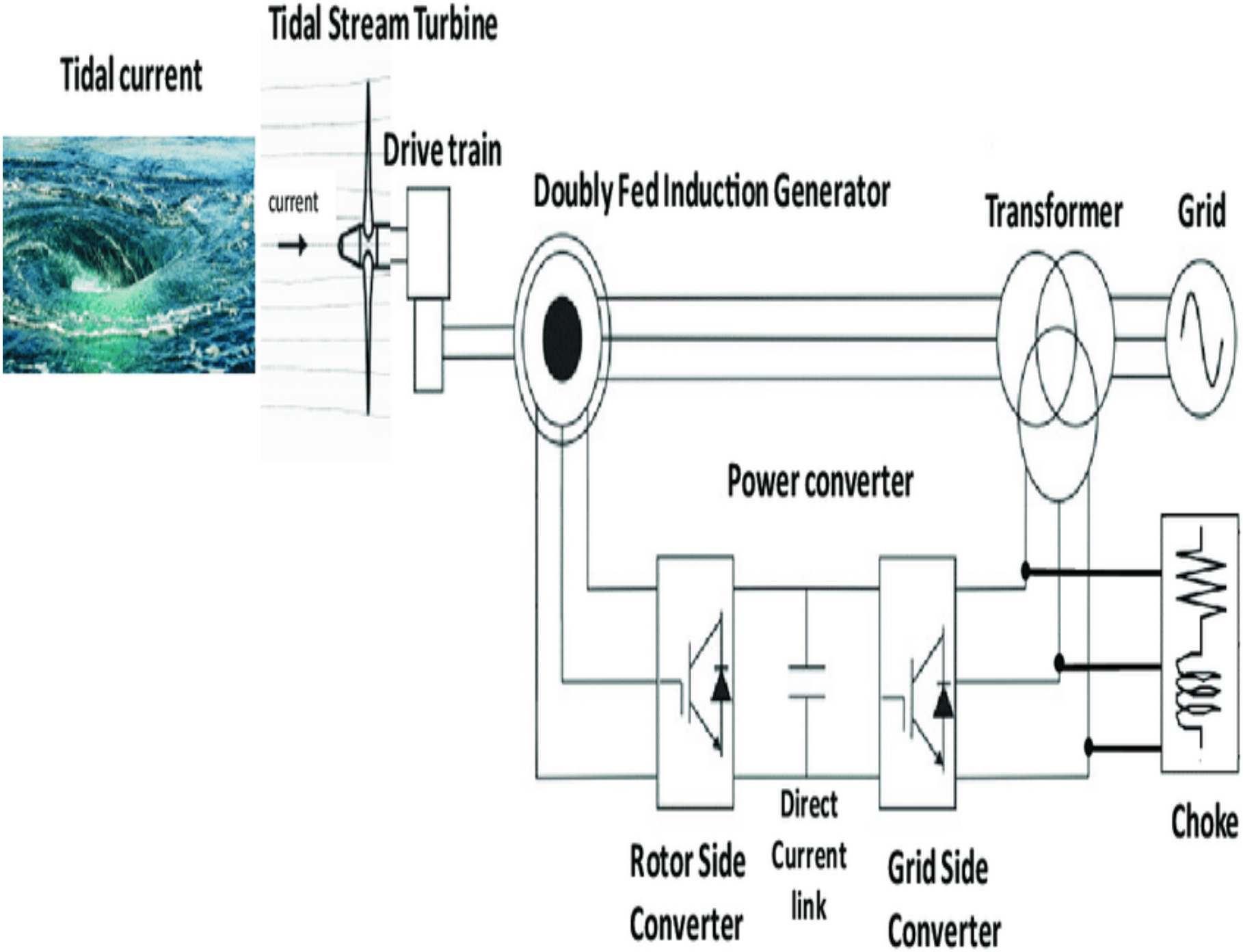
- https://www.irena.org/documentdownloads/publications/tidal_energy_v4_web.pdf

Types of Tidal Energy Generation

- Tidal Barrage
- https://energyeducation.ca/encyclopedia/Tidal_barrage
  - Ebb generation
  - Two way generation

Tidal Stream Generator

- A tidal stream generator, also known as a tidal energy converter or tidal stream turbine, is a device that harnesses the kinetic energy of moving water, specifically tides, to generate electricity, similar to how wind turbines use wind.


Tidal Stream Turbine
Tidal current
Drive train
current
Doubly Fed Induction Generator
Transformer
Grid
Power converter
Direct
Current
link
Rotor Side
Converter
Grid Side
Converter
Choke

Barrage Logistics

- Largest surface area, greatest tidal range
  - Square Effect
- Effects of ecosystem (fish, sediment, etc.)
- Proximity to population

Historical Aspects

Ancient Tide Mills

- During the Roman occupation of England, tide mills were built to grind grain and corn
- These tide mills operated by storing water behind a dam during high tide.
- As the tide receded the water was slowly let out from behind the dam in order to power the mill.

Location

- Tidal mills were usually built on inlets branching off tidal estuaries.
- An estuary is a wide part of a river where it meets the sea.
- Tidal estuaries are characterized by narrow, shallow channels with a relatively constant width and depth.
- Tides are greatly amplified in these areas of smaller volume, which causes the tide to travel up the river.

Tidal Barrage

- Similar to a dam
- Structure must span the entire width and height of the estuary

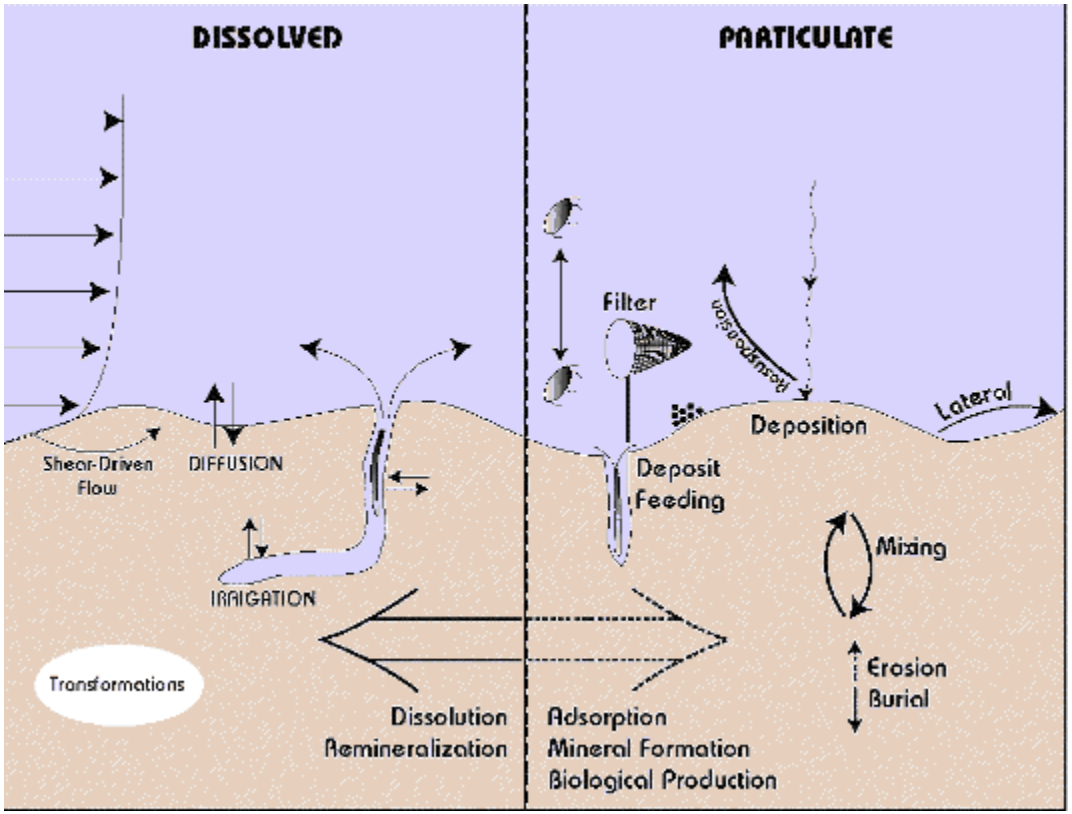
Advancement in the Tide

Creating a new paradigm for renewable energy choices

Impacts

- Environmental
- Economic
- Social

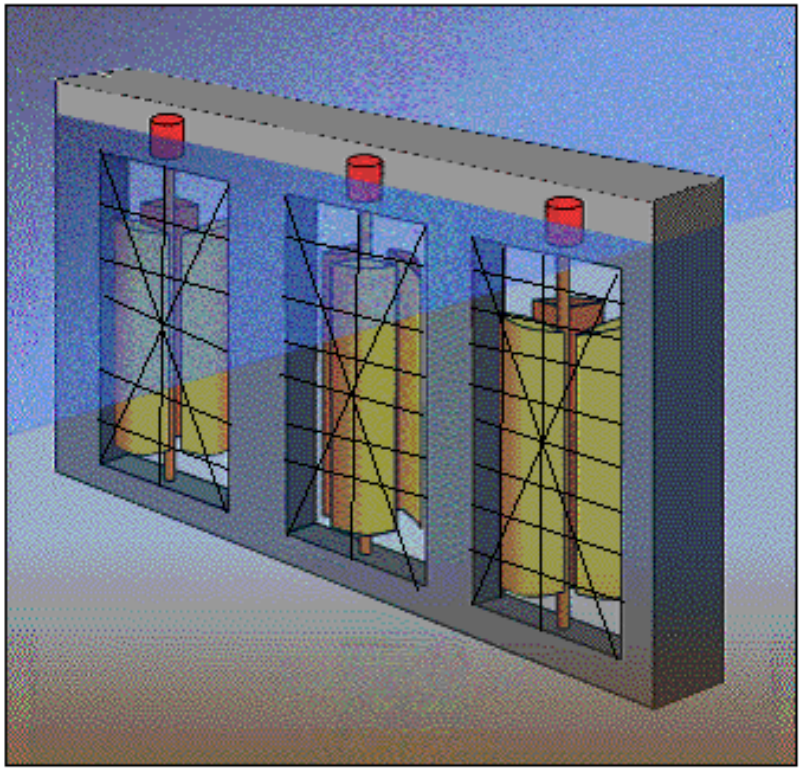
Boundary Layer Effects



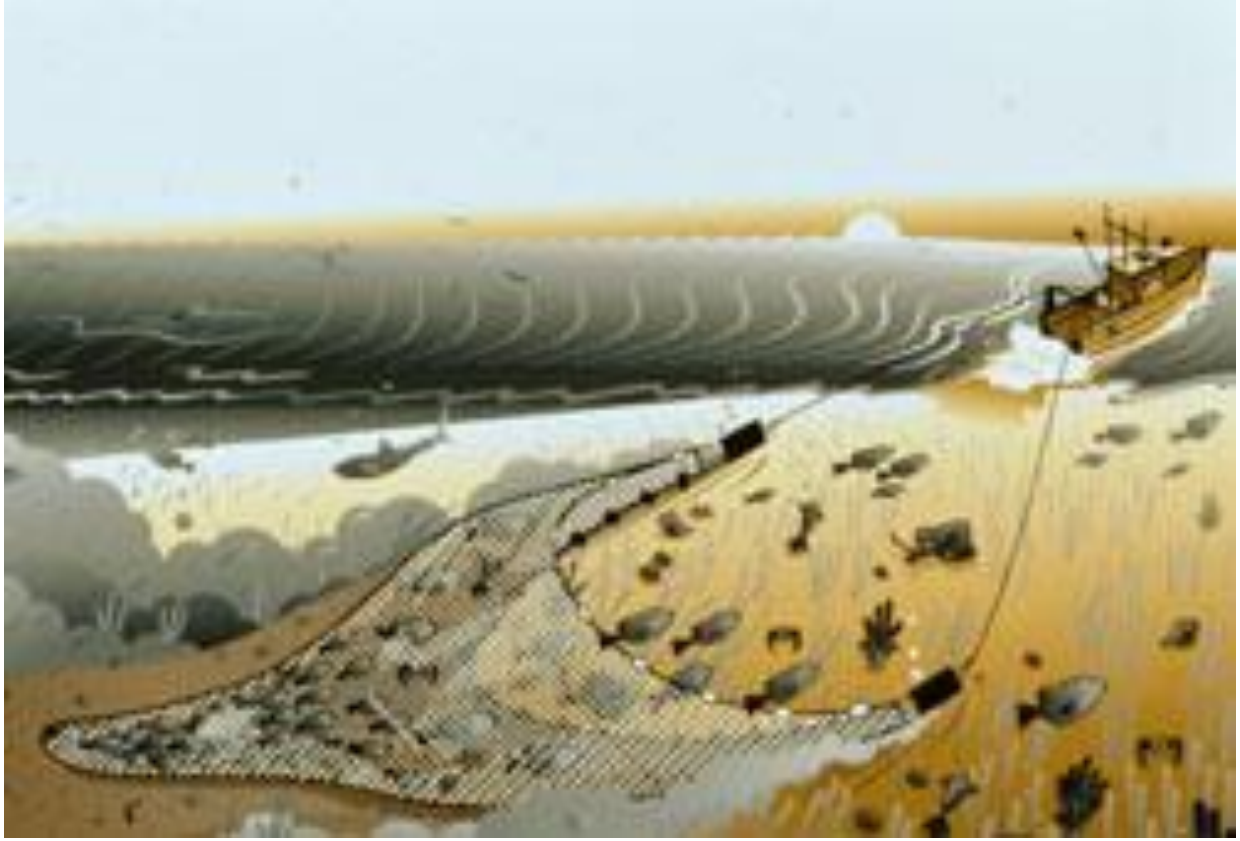
http://129.173.3.163/BBLCh1.html

A More Futuristic Approach – Artificial Reef Program: Offshore Rigs

Marine Conservation Structure



Maximized Conservation



Conclusions

- Tidal Energy is interesting!
- Economics are difficult to get around
- These projects are expensive
- Are people really willing to pay?